FACULTY OF COMPUTING

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SECP3744 – ENTERPRISE SYSTEMS DESIGN AND MODELLING (WBL)

SECTION 1

TITLE:

ENTERPRISE SYSTEM DEVELOPMENT AND MANAGEMENT (ESDM) PROJECT

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TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Universiti Tun Hussein Onn Malaysia (UTHM) is a public higher education institution that manages diverse academic and administrative activities to support students, academic staff, and university management. These functions are supported by several core operational information systems that collectively form the university's enterprise systems environment. As the scale and complexity of university operations increase, the need for structured coordination between business processes, information systems, and supporting technologies becomes more significant.

Currently, UTHM utilizes a basic form of enterprise architecture to support its information systems landscape. While this provides a general structure for system usage and management, it does not yet represent a comprehensive or formally defined enterprise architecture that fully aligns business needs with application, data, and technology layers. As a result, there is an opportunity to further develop and formalize enterprise architecture to improve system integration, consistency, and long-term planning.

This project involves the development of a proposed enterprise architecture for UTHM that covers four core operational systems at the enterprise level. Each system is analyzed by a different project group to provide detailed architectural insights. The focus of this project is on Student Information System (SIS), with particular emphasis on the Student Course Registration subsystem, By applying enterprise architecture principles commonly used in higher education institutions, this project aims to propose a structured architectural model that supports clearer system alignment, improved information flow, and more effective management of student academic processes at UTHM.

1.2 PROBLEM STATEMENT

This section outlines the key problems that motivate the development of the proposed enterprise architecture for UTHM. It highlights current issues related to the existing enterprise architecture, associated inefficiencies, and potential risks affecting student-related systems. The discussion focuses on the Student Information System (SIS), particularly the Student Course Registration subsystem, to justify the need for a structured enterprise architecture approach.

1.2.1 Current Issues

UTHM utilizes multiple information systems to support its academic and administrative operations, including student related systems such as SIS. However, the enterprise architecture currently applied is basic and not formally documented as a comprehensive enterprise-wide model. As a result, the relationships between business processes, applications, data, and technology are not clearly defined across the university's core operational systems.

1.2.2 Inefficiencies

Due to the absence of a formalized and standardized enterprise architecture, coordination between systems may be inconsistent. System development and enhancement activities may be carried out independently by different units, leading to challenges in system integration and information flow. These inefficiencies are more apparent in student-related systems, where accurate data handling and process alignment are critical to supporting academic activities such as course registration.

1.2.3 Risks and Limitations

Without a clear architectural structure, the Student Course Registration subsystems under the SIS faces limitations in terms of scalability, and long-term planning. The lack of clear mapping between business requirements and system functions increases the risk of misalignment with UTHM broader academic and digital objectives. In addition, future system improvements may become more complex and harder to manage without a structured enterprise architecture to guide decision making.

1.3 PROJECT OBJECTIVES

The main objective for this project is to design and purpose an enterprise architecture for Universiti Tun Hussein Onn Malaysia (UTHM). The objectives will be focused on analyzing the current system environment, applying a structured enterprise architecture framework which is TOGAF framework, and supporting future enhancing management of university operations. The objectives of this project are as follows:

1. To analyze the existing information system used by Universiti Tun Hussein Onn Malaysia (UTHM) including academic system, management system and others to gain greater understanding the current system environment and challenges.

2. To design an Enterprise Architecture (EA) for UTHM by using TOGAF framework as a structured guideline to structure the business processes, applications, data, and technology

3. To provide a structured architectural reference for future system improvement to make the system more manageable and systematic

1.4 PROJECT SCOPE

This section defines the project boundaries to ensure that this project is achievable within the given constraints. It clearly outlines the areas that covered by the project (in-scope) and the part that are not included (out-of-scope) particularly related to the proposed enterprise architecture for UTHM.

1.4.1 IN-SCOPE

The in-scope areas of this project include:

- Analysis the existing enterprise systems environment at UTHM such as overview of academic, student, and management-related system.

- Focus on the Student Information System (SIS) specifically the student Course Registration subsystem as main case study

- Design a proposed Enterprise Architecture for UTHM using the TOGAF framework as structured guideline that are covering key architectural layer such as Business, Application, Data and Technology Architecture

- Create an architectural models and documentation as a reference for improvement of the system UTHM in future.

1.4.2 OUT-OF-SCOPE

The out-of-scope areas of this project include:

- Development a system, coding or any related to implementation of proposed enterprise architecture.

- Perform a testing such as security testing and user acceptance testing of existing or proposed system.

- Evaluation of hardware, cost analysis, or budgeting related to system implementation.

1.5 PROJECT IMPORTANCE

The development of a formalized Enterprise Architecture (EA) for UTHM is critical to ensuring the university's digital infrastructure can support its growing academic and operational demands. This project holds significant importance in three key areas:

1.5.1 Strategic Alignment and Resources Optimization

Currently, UTHM's information systems operate in silos, which often leads to redundant spending where different faculties might procure separate software solutions for similar needs. This project is important because it utilizes the TOGAF framework to create a unified blueprint. By centralizing the architecture, the university can identify overlapping resources and streamline IT investments. This ensures that every ringgit spent on technology aligns directly with UTHM's strategic objectives, reducing unnecessary costs, and fostering long-term sustainability.

1.5.2 Operational Efficiency and System Integration

The project addresses the critical issue of fragmentation within UTHM's current environment. By proposing a structured architecture for the Student Information System (SIS), this project eliminates the inefficiencies caused by independent system development. It provides a standard reference for IT teams, ensuring that data flows seamlessly between the academic and administrative departments. This reduces data redundancy, minimizes the risk of integration errors, and simplifies the maintenance of complex university systems.

1.5.3 Enhanced Student Experience

Since the project focuses specifically on the Student Course Registration subsystem, its immediate importance lies in directly improving student experience. Course registration is a high-traffic process that frequently suffers from system congestion. A well-architected system ensures scalability, preventing system crashes during peak registration times. Furthermore, this architectural update aims to streamline the user interface, making the registration process more intuitive and less stressful for students.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

This chapter presents a literature review related to enterprise architecture (EA) in universities, providing both theoretical and practical foundations for this study. This chapter begins with EA definition, key concepts, and some applications within higher education institutions. Meutia, Sulistiani, Budiarti, and Sari (2022) state that EA serves as a strategic framework that aligns business processes, information system, and technology infrastructure with organizational objectives, helping universities to manage complex operations and support digital transformation.

Besides, the review also highlights previous studies on EA in universities, emphasizing how EA frameworks have been implemented in universities to coordinate administrative, academic, and technical functions (Hindarto, 2023; Amin, Hussain, Daoud, Abu Bakar, & Ahmad, 2024). In addition, this chapter examines various subsystems, discussing the existing implementation and architectural considerations of these sub-systems. Furthermore, this chapter also explores the technologies used in EA and its sub-systems, including software platforms and architectural frameworks.

Finally, this chapter concludes with a summary of key findings from previous studies to establish a solid foundation for understanding EA in universities worldwide.

2.2 ENTERPRISE ARCHITECTURE

Enterprise Architecture (EA) is a method that an organization uses to plan the business and IT in a way that each will fit together well. Following the TOGAF standard, EA segments the organization into different segments to show how processes, information, applications, and technology support each other (The Open Group, 2018). EA also serves as a clear blueprint to map business goals to their supporting IT systems (Lankhorst, 2017). EA typically includes four sectors which are Business Architecture, Data Architecture, Application Architecture, and Technology

Architecture (The Open Group, 2018). Together, it helps everyone understand how the organization works and how IT supports the organization.

In University, the Data Architecture layer is crucial, as they require high data quality and data organization for all aspects of their business, including students, staff, and research. Previous works indicate that universities often use a TOGAF framework for guidance, but they adjust it to their needs (Pedroso, 2021). Universities focus on management of organizational data, interoperability of organizational systems, accessibility, and security. However, challenges like different departments managing their own systems, legacy and outdated technologies, and a lack of trained EA professionals prevent them from implementing EA effectively. Thus, universities require good planning and teamwork for the successful application of EA.

2.2.1 UNIVERSITY OF LAMPUNG

A study by Nama and Kurniawan (2017) designed an adaptive IT infrastructure based on the enterprise architecture framework The Open Group Architecture Framework (TOGAF) Architecture Development Method (ADM) for the University of Lampung (Unila). The Academic Information System (Siakad) used by Unila, which has been operational for a long time, is not integrated with other applications such as finance, employee, and planning systems. This causes delays in gathering data when leaders or stakeholders request reports across the work unit.

Figure 2.1 shows the enterprise architect that was proposed to Unila. According to the study, the IT unit should be in charge of centralizing infrastructure management. To deliver a scalable, dependable, and flexible service, the unit must also implement security rules, data management and distributed system services and run the Disaster Recovery Center.

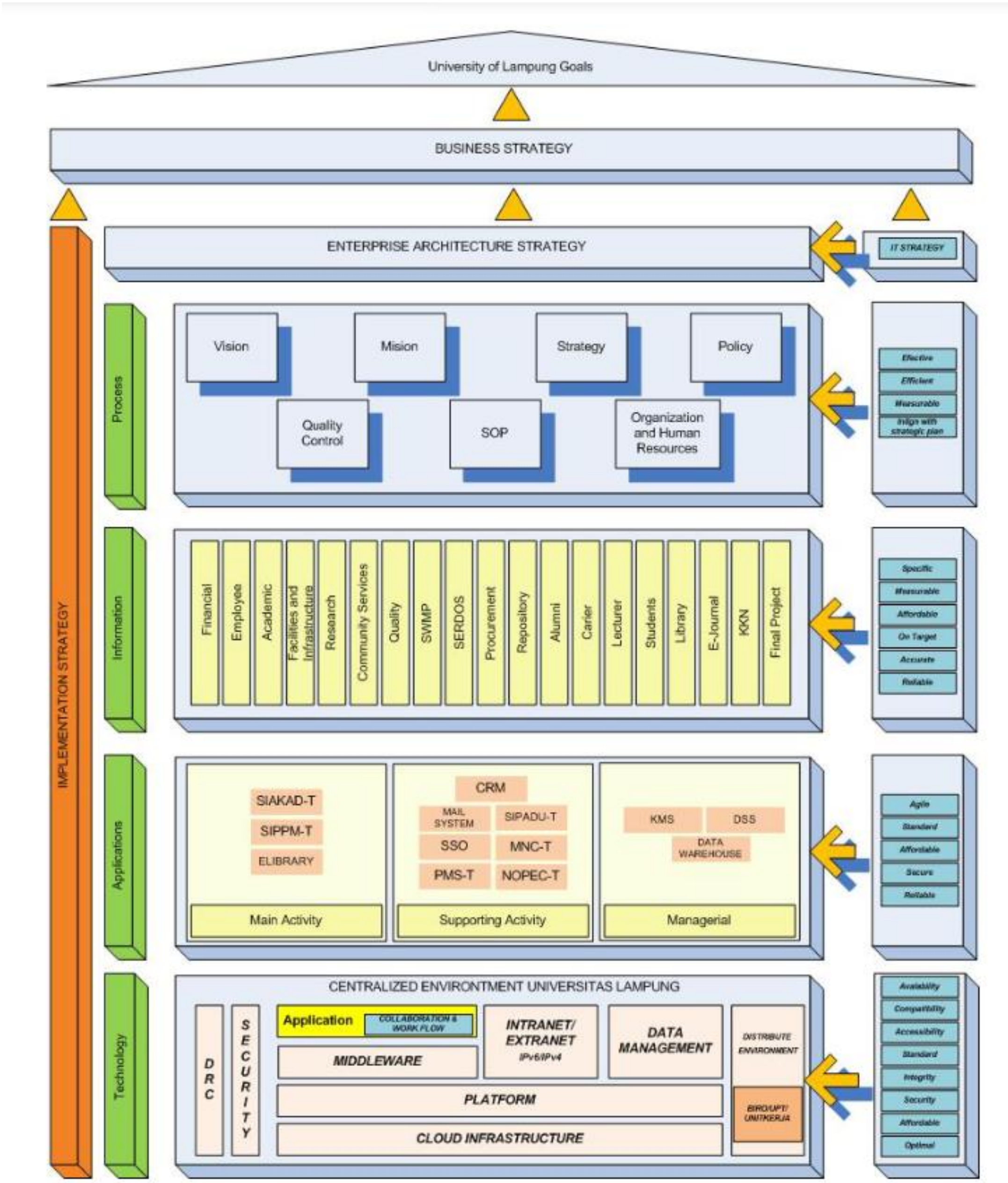

Figure 2.1: Enterprise Architecture proposed to Unila. (Nama & Kurniawan, 2023)

2.2.2 NATIONAL UNIVERSITY SOUTH JAKARTA

Another study by Hidarto (2023) investigates the importance of EA in enabling digital transformation of the management systems in universities. The paper uses The Open Group Architecture Framework (TOGAF) framework as its research method, as it provides a methodical framework for the development, organization, execution, and oversight of enterprise information technology architecture. The paper defined digital transformation as the migration from manual processes to using application systems that can automate those processes, as universities often face problems such as human errors and slow information flow and data retrieval from manual processes. The paper also highlights opportunities using application systems such as access to advanced technologies, artificial intelligence (AI), big data analytics, and the Internet of Things (IoT) that can increase effectiveness in operation, boost the process of teaching and use resources efficiently.

Figure 2.2 shows the enterprise architecture described by the paper. The Integration Services establishes a connection to multiple sources, allowing data integration from external systems using API. The portal is a central interface through which teachers, staff, and students can combine their access to information and university resources. Data & Analytics oversees collecting, analysing, and using data to better decision-making, while the Back-Office systems manage the internal operations of the university.

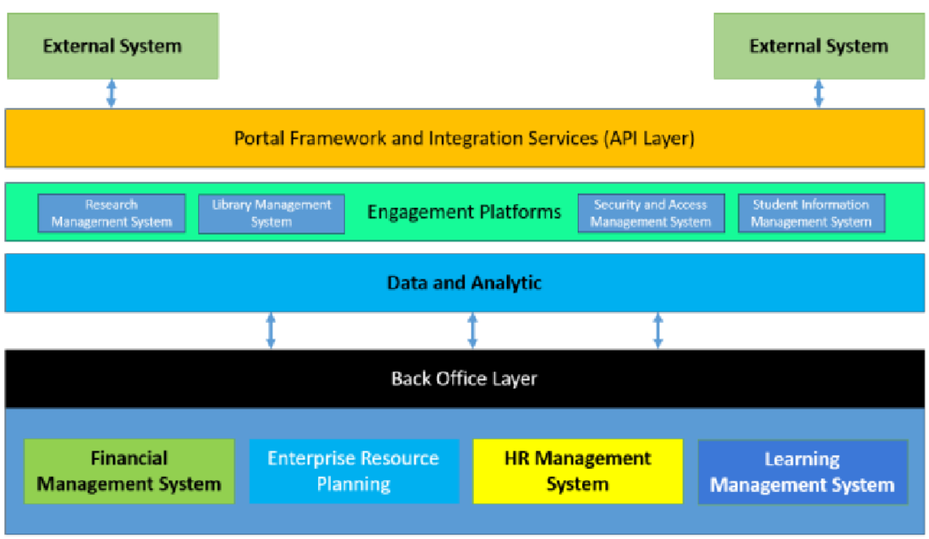

Figure 2.2: Enterprise Architecture by Hidarto. Hidarto (2023)

2.2.3 New York University

Figure 2.3 shows the New York University's (NYU) Enterprise Architecture operating model where it explains how the university connects its overall strategy with IT planning and governance. NYU uses EA to connect key areas such as learning and teaching, research, community engagement, and administration with its IT and digital systems. This shows that EA is not only about technology, but it also about supporting the university's main goals. By aligning EA with NYU's strategic, the university ensures that IT investments and digital initiatives support teaching, research, and daily operations across different campus.

At the architecture level, NYU's EA covers business process, data, application and technology infrastructure. Important data such as student records, research information, and financial data are managed clearly to support reporting, better decision making, etc. Core system like the Student Information System (SIS), Learning Management System (LMS), and identity services are shared across the university to avoid any duplication in the system. In addition, technology standards such as cloud, security governance, and infrastructure guidelines help support both academic and research activity. Overall, New York University Enterprise Architecture shows how strong governance can help universities manage a complex system while keeping IT decisions match with institutional goals, making it useful reference for other universities.

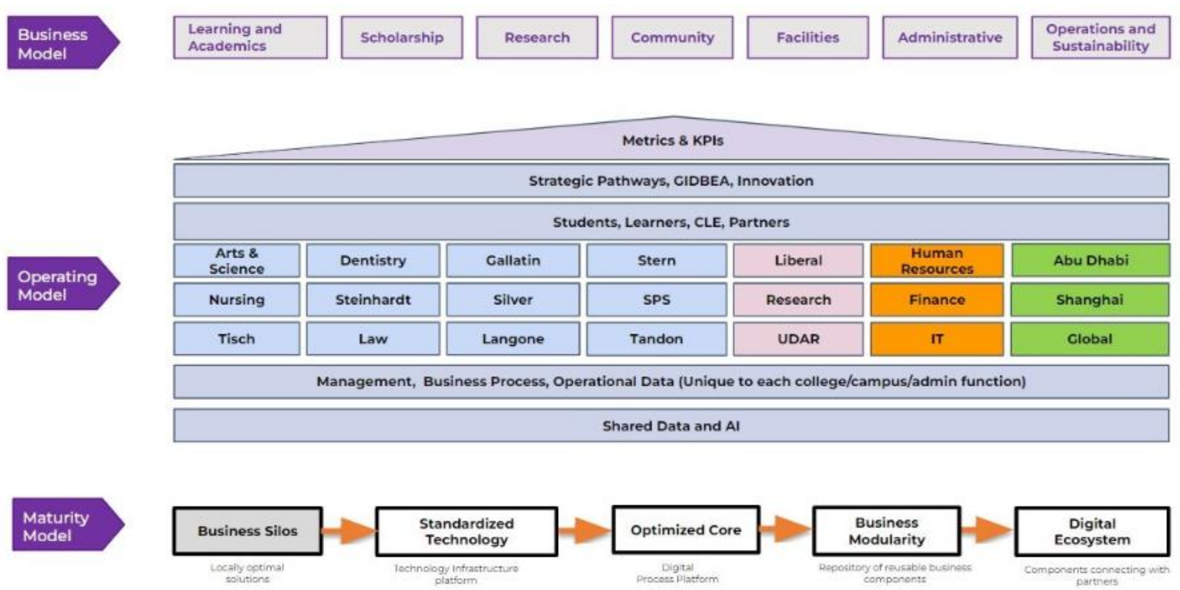

Figure 2.3: Enterprise Architecture New York University

2.3 RELATED SUB-SYSTEM

Research has been done to identify the subsystems used in university by analyzing them through the types of Enterprise Architecture methods and technologies. These studies look at the implementation and integration of systems like the Human Resources Management System (HRMS), Academic Information System (AIS), Student Information & Subject Registration System (SIS) and Learning Management System (LMS) and Business Intelligence (BI) and Data within EA frameworks to support coordinated processes, data consistency and higher education institutions' operational and strategic goals.

2.3.1 Human Resources Management System (HRMS)

An HRMS is a system that combines information technology and human resources management (HRM) to automate and support HR procedures. For instance, it manages payroll, training, performance reviews, resignations, resume (applicant), employee details and attendance. The goal is to increase operational efficiency in HR departments, decrease manual work and make data access more dependable and simpler (Navaz et. al, 2013).

According to Sikarwa, 2018, the primary goal of HRMS is to guarantee that the organization's strategic goals can be met and the universities' needs can be satisfied in terms of both quantity and quality. The method used from the study is the Open Group Architecture Framework (TOGAF) 9.1 where it stated at preliminary phase; in addition to managing the standard data sources in universities like payroll, employee attendance, hiring and leave, HRMS also handles the demands of promotion management, period payment increase analysis, reward and punishment management and staff education. In particular, HRMS in universities mainly covers these functional areas including payroll management, human resources management, institutional management, civil service management, education and training management, performance management and function management. Furthermore, universities have a HRMS integrated system design that can be integrated with current systems so that its HRMS architecture may supply information required by end users as well as the business needs of universities in managing human resources.

2.3.2 Academic Information System (AIS)

A university's Academic Information System (AIS) is a software program that keeps track of all academic data including timetables, grades, courses and student information. Students can access their transcripts, study result cards and academic information online while administrators can effectively manage student records, enter grades, manage courses and create reports. It replaces manual, paper-based procedures, saves time, minimizes mistakes and improves staff and student access to academic management (Manuhutu et. al, 2018).

According to Astri and Gaol, 2013, the academic information system manages the school's academic processes from new student entrance to graduation ceremonies. In the study, this system performs essential functions such as managing admission test results, scheduling, curriculum, grade management, student administration and academic report applications. The study uses the Enterprise Architecture Planning (EAP) methodology to connect technology, data and apps across the systems. Hence, better administration, coordination and long-term system planning are made possible by the school's ability to create a comprehensive blueprint through EAP that synchronizes its academic information systems with an integrated enterprise-level architecture.

2.3.3 Student Information & Subject Registration System (SIS) and Learning Management System (LMS)

The Student Information System (SIS) and the Learning Management System (LMS) constitute the core technological foundation that supports both academic administration and instructional delivery within a university. In contemporary digital transformation initiatives these systems are expected to function not as isolated components but as fully integrated parts of a comprehensive University Management System (UMS) (Hindarto, 2023).

The SIS is responsible for handling the complete lifecycle of student information. This includes initial registration, enrollment in subjects and the maintenance of academic records, and the management of graduation processes. Earlier forms of SIS frequently encountered issues such as fragmented datasets and duplicated entries which resulted in inconsistencies across institutional records (Lamey et al., 2023). To address these challenges many institutions, rely on EA frameworks that are often modelled on methodologies such as TOGAF. These frameworks enable the design of more coherent processes, unified databases, and stronger linkages to subsystems that include finance and scheduling (Metia et al., 2022).

The LMS complements the SIS by managing instructional resources and assessments. Effective academic operations require that the LMS maintain seamless integration with the SIS to ensure the accurate synchronization of enrollment data and student performance records. Lamey et al. (2023) emphasize that the development of intelligent and integrated academic solutions depends on consistent and harmonized data shared across both systems. Case studies demonstrate that EA planning (Nama and Kurniawan, 2017) plays a central role in shaping the Application Architecture layers to support reliable academic data flows.

2.3.4 Business Intelligence (BI) and Data

The Business Intelligence (BI) and Data Subsystems represent the strategic dimension of Enterprise Architecture implementation. Its purpose is to transform institutional data into reliable insights that guide decision making, policy information, and overall governance. A central objective of EA within digital transformation initiatives is to advance universities toward a fully data driven operational environment (Hindarto, 2023).

The effectiveness of any BI subsystem is determined by the strength of the Data Architecture established during the EA planning process. This stage involves the standardization of data definitions and the assurance of data quality throughout all primary source systems such as the SIS, HRMS, and FMS before these data are transferred into a centralized data warehouse (Meutia et al., 2022). Without rigorous data governance the outputs produced by BI platform risk becoming fragmented or unreliable.

According to Lamey et al. (2023) advanced analytical and intelligent institutional systems require a well-integrated and dependable data foundation. Such a foundation allows universities to perform tasks that include forecasting student retention, allocating resources efficiently, and monitoring institutional performance indicators. Consequently, the application of EA principles (Nama and Kurniawan, 2017) extends beyond merely connecting technological subsystems. It involves structuring the data environment in a way that supports meaningful strategic intelligence.

2.4 TECHNOLOGY USED

This section serves to describe and justify the key technological platforms, frameworks, and tools selected for the implementation of the enterprise system under study. This part highlights the infrastructure and digital solutions chosen for their relevance, effectiveness, or innovation, providing a clear rationale for why these technologies suit the project's goals and context. In this case of university enterprise systems, this section explains how foundational technologies like SAP

Business Technology Platform (BTP) support the digital transformation of campus services and operations, allowing for centralized management, real time analytics, and seamless integration across diverse academic and administrative functions.

2.4.1 King Abdullah University of Science and Technology (KAUST)

The project uses the SAP Business Technology Platform as a central technological enabler, similar to the real-life implementation at King Abdullah University of Science and Technology. Building on SAP BTP, KAUST created an integrated digital platform called "KAUST Central," which consolidated the essential campus services-dining, transportation, security, payments, and communication-into one mobile and web application. The system is exemplary, depicting how SAP BTP can enable a university-wide digital transformation through scalable cloud services, intelligent integration, and user-centered design.

Several SAP BTP components were leveraged for the KAUST project. SAP Mobile Services SDK was used to build native apps for iOS and Android, thus allowing real-time access to campus functions. SAP Analytics Cloud allowed data-driven decisions based on tracking student behavior and providing personalized content, while SAP Experience Management (Qualtrics) continuously gathered user feedback for refining campus services. Moreover, the SAP BTP cloud infrastructure enabled seamless integrations with legacy systems and third-party applications through standardized APIs, which is very important in a distributed university setting where various departments operate on different platforms.

From an enterprise architecture perspective, with SAP BTP, KAUST has implemented a modular and flexible architecture scaling across departments while preserving centralized governance. Single sign-on simplified access to the system, while the cloud-native setup provided support for rapid updates and extensions, such as adding COVID-19 information features and smart city modules. Most impressively, the platform has managed to cut campus services' wait times from 10 minutes down to 40 seconds and allowed logistics, dining, and transport on campus to be coordinated through real-time dashboards and automated processes.

The use of SAP BTP at KAUST represents one way in which universities are moving beyond traditional ERP models toward a campus-as-a-platform model. It confirms SAP BTP as appropriate, future-ready technology for university enterprise systems, which supports digital innovation, integration, analytics, and smart campus ecosystems.

2.5 SUMMARY

The literature review highlights the importance of Enterprise Architecture (EA) in helping universities to align their business procedures, data management, applications, and technology infrastructure to meet institutional goals and digital transformation. Previous studies show that universities often adopt EA frameworks such as TOGAF framework, but universities usually adjust the framework so that they could fit their own needs, especially when dealing with issues such as decentralized departments, out-dated systems, and diverse stakeholder needs.

Case examples from University of Lampung demonstrate how EA can support better alignment between strategy and IT, improve data management, integrate applications, and set clear technology standards. Meanwhile, insights from the National University South Jakarta emphasize that universities often change from manual processes to digital transformation using application systems that can automate those processes.

In addition, the review of key university sub-systems, including HRMS, AIS, SIS, LMS, BI, and Data shows how EA helps universities ensure smooth data flow, reduce duplication, and design scalable systems. Furthermore, the use of technologies, such as SAP Business Technology Platform (BTP) as demonstrated by KAUST, further show how cloud-based platforms enable universities to deliver smart campus services and support long-term digital innovation.

In conclusion, the literature establishes EA as a strong foundation for modern university enterprise systems, providing both strategic direction and technology integration to enhance overall institutional performance.

CHAPTER 3

METHODOLOGY

3.1 INTRODUCTION

This project aims to design an enterprise architecture for Universiti Tun Hussein Onn Malaysia (UTHM). To achieve this goal, the Waterfall development approach is adopted in this project. The methodology guides the development process of this project from starting to the end, which is planning, analysis, design, implementation, and testing the enterprise architecture components. By following the methodology, the project ensures that every phase of the methodology was implemented successfully and aligned with UTHM operational requirements.

3.2 CHOSEN METHODOLOGY

The Waterfall methodology is a sequential development process where every phase must be completed before moving on to the next phase. It is suitable to be applied in projects that have clear and fixed requirements.

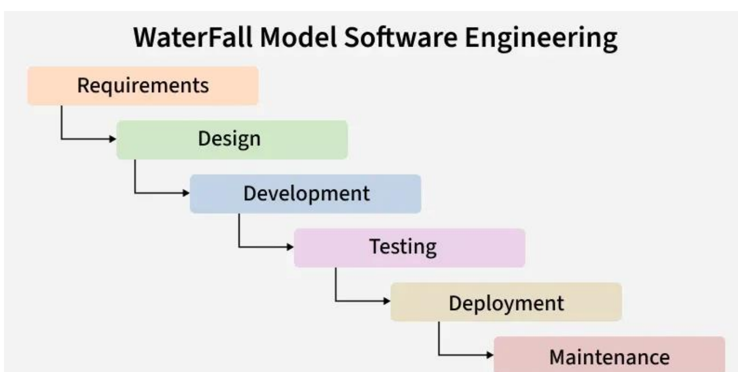

Figure 3.1 Waterfall Methodology Phases (GeeksforGeeks, 2025)

The reason why Waterfall methodology is chosen is because of its structured flow. Since enterprise architecture involves complex systems and long-term planning, understanding the correct requirements is very important. With this methodology, all requirements must be clearly understood before moving to next phase to design the real enterprise architecture.

Other than that, each phase in the Waterfall approach such as requirements analysis, design, implementation, and testing is clearly documented. These documentations will help the university a lot in future maintenance or enhancement.

Moreover, since the phases are clearly defined, it is easier to manage resources, manpower, and project schedules. This reduces uncertainty of the project and reduces the risk of project failure. Therefore, the Waterfall methodology is a suitable approach for this project.

3.3 PHASES OF METHODOLOGY

As illustrated in Figure 3.1, this project is led by the six distinct phases of the Waterfall Model. Each phase depends on the deliverables of the previous one, making sure of a logical progression from recognizing the fragmentation issues at UTHM to delivering an interrelated Enterprise Architecture blueprint.

3.3.1 Requirements

The first phase involves gathering and analysing the necessary information to understand the current state of UTHM's business and IT environment specifically identifying pain points caused by the separation of various departmental systems. In this stage, the project team conducts data gathering to understand the existing siloed systems across different faculties and administrative units, defining the scope for unifying these departments into a single platform. The outcome of this phrase is a comprehensive Problem Statement and User Requirement Specification (URS) that clearly identifies the gaps in the current architecture.

3.3.2 Design

Based on the requirements gathered, the team moves to the design phase to create the architectural blueprints that define the "To-Be" state of the organization. The team adopts the TOGAF (The Open Group Architecture Framework) approach to model the target architecture including the Business, Data, Application, and Technology layers required to support a unified university platform. This phase concludes with the production of high-level architectural diagrams and a proposed Enterprise Architecture design document that solves the issues identified in the analysis phase. technologies were defined in this phase in order to design the system architecture, which was done by using SAP Build Apps. Each member was assigned different modules and was assigned to make visual representations of the modules using UML diagrams such as use case diagrams, ERD diagrams, use case descriptions, activity diagrams, and sequence diagrams. Interface design was also provided to give an overview of how the finished product would look.

3.3.3 Development

During the development phase, the system was constructed with SAP Build Apps, and the process began with a structured implementation of the data layer, building the necessary entities based on each member module. Once the data foundation was established, UI design and logic implementation were completed, with functional modules customized to each module.

3.3.4 Testing

This phase focused on testing the quality of the custom logic and data bindings, beginning with unit testing to ensure that the complex lookup formulae for dynamic session filtering produced exact data from the database. Functional testing was then performed using the SAP Build Apps Previewer, where each use case module was tested to ensure that the system appropriately handled both success and failure scenarios, such as mismatched passwords or wrong current credentials. Finally, integration testing verified that the data layer and user interface stayed in sync, guaranteeing that changes to one page of the system's information were promptly reflected in the application's local memory. The predicted and actual results were then compared to ensure that all functional requirements were properly met.

3.4 PROJECT PLANNING SCHEDULE

The project planning schedule, as illustrated in Figure 3.2, follows the sequential stages of the Waterfall methodology to ensure a structure development process. The timeline begins with a collective planning and analysis phase, where the entire team conducts a site visit to UTHM and prepares interview questions to define the system requirements. The project moves into design phase, where individual responsibilities are assigned to all team members to develop specific tasks for the seven modules. The schedule then transitions into the implementation phase, focusing on database creation, and coding. Finally, the process concludes with a testing and deployment phase.

TASKS	ASSIGNED TO	DURATION (WEEKS)	27/10-31/10	3/11-7/11	10/11-14/11	17/11-21/11	1/12-5/12	8/12-12/12	15/12-19/12	22/12-26/12	29/12-2/1	5/1-9/1	12/1-16/1
Meeting with group members	ALL	1											
Project planning	ALL	1											
Requirements													
Overview and Problem Statement	SYAHMI	2											
Project Objectives and Project Scope	NAIM	2											
Project Important	ADRIANA	2											
Literature Review, EA concepts, explore SAP BTP	ALL	2											
Analysis													
Prepare Interview Questions	ALL	1											
Site Visit to UTHM	ALL	1											
Define System Requirements	FARRA	1											
Company Organization Structure	SAFIYA	1											
Analyis Current System	CHUA	1											
Design													
EA AS-IS & TO-BE	ALL	2											
System Architecture	IMAN	2											
Use Case Diagram and Description													
Process 1 : View Student List for each Section Module	FARRA	1											
Process 2 : Credit Hour Module	NAIM	1											
Process 3 : Section Scheduling Module	ADRIANA	1											
Process 4: Student Course Registration Module	CHUA	1											
Process 5: Manage Student Profile Module	IMAN	1											
Process 6 : Section Allocation Module	SYAHMI	1											
Process 7: Course Approval Module	SAFIYA	1											
Activity diagram													
Process 1 : View Student List for each Section Module	FARRA	1											
Process 2 : Credit Hour Module	NAIM	1											
Process 3 : Section Scheduling Module	ADRIANA	1											
Process 4: Student Course Registration Module	CHUA	1											
Process 5: Manage Student Profile Module	IMAN	1											
Process 6 : Section Allocation Module	SYAHMI	1											
Process 7: Course Approval Module	SAFIYA	1											
Sequence diagram													
Process 1 : View Student List for each Section Module	FARRA	1											
Process 2 : Credit Hour Module	NAIM	1											
Process 3 : Section Scheduling Module	ADRIANA	1											
Process 4: Student Course Registration Module	CHUA	1											
Process 5: Manage Student Profile Module	IMAN	1											
Process 6 : Section Allocation Module	SYAHMI	1											
Process 7: Course Approval Module	SAFIYA	1											
Database Design													
Process 1 : View Student List for each Section Module	FARRA	1											
Process 2 : Credit Hour Module	NAIM	1											
Process 3 : Section Scheduling Module	ADRIANA	1											
Process 4: Student Course Registration Module	CHUA	1											
Process 5: Manage Student Profile Module	IMAN	1											
Process 6 : Section Allocation Module	SYAHMI	1											
Process 7: Course Approval Module	SAFIYA	1											
Interface Design													
Process 1 : View Student List for each Section Module	FARRA	1											
Process 2 : Credit Hour Module	NAIM	1											
Process 3 : Section Scheduling Module	ADRIANA	1											
Process 4: Student Course Registration Module	CHUA	1											
Process 5: Manage Student Profile Module	IMAN	1											
Process 6 : Section Allocation Module	SYAHMI	1											
Process 7: Course Approval Module	SAFIYA	1											
Development													
Create Database													
Process 1 : View Student List for each Section Module	FARRA	2											
Process 2 : Credit Hour Module	NAIM	2											
Process 3 : Section Scheduling Module	ADRIANA	2											
Process 4: Student Course Registration Module	CHUA	2											
Process 5: Manage Student Profile Module	IMAN	2											
Process 6 : Section Allocation Module	SYAHMI	2											
Process 7: Course Approval Module	SAFIYA	2											
Coding of system's main function													
Process 1 : View Student List for each Section Module	FARRA	2											
Process 2 : Credit Hour Module	NAIM	2											
Process 3 : Section Scheduling Module	ADRIANA	2											
Process 4: Student Course Registration Module	CHUA	2											
Process 5: Manage Student Profile Module	IMAN	2											
Process 6 : Section Allocation Module	SYAHMI	2											
Process 7: Course Approval Module	SAFIYA	2											
Testing & Deployment													
EA Presentation	ALL	1											
Final Project Demo	ALL	1											

Figure 3.2: Gantt Chart

3.5 SUMMARY

The enterprise architecture for Universiti Tun Hussein Onn Malaysia (UTHM) is developed using the Waterfall methodology, which follows a structured and sequential development process. This approach organizes the project into clear phases, including planning, analysis, design, implementation, and testing, ensuring that each stage is completed before moving forward and remains aligned with UTHM operational requirements.

The Waterfall methodology is suitable for this project because enterprise architecture involves complex systems, and long-term planning that require clearly defined requirements. Understanding all requirements at the early stage helps reduce the risk of system misalignment and improves the quality of architectural design. In addition, clear documentation produced in each phase supports future system maintenance and enhancement activities.

The project planning schedule follows the sequential flow of the Waterfall model, beginning with site visits and requirement analysis. The process then moves into the design phase, where responsibilities are distributed among team members, followed by implementation activities such as database creation and the use of framework such SAP Build. The project concludes with testing and deployment to ensure the proposed enterprise architecture functions as intended within the university environment.

CHAPTER 4

ANALYSIS AND DESIGN

4.1 INTRODUCTION

This chapter's focuses on the design and analysis of the proposed system and the Enterprise Architecture (EA) for Universiti Tun Hussein Onn Malaysia (UTHM). It looks at the current system to find issues and requirements followed by the right enterprise architecture and system design approaches to create an improved EA. This chapter's outputs include the system architecture, database design and interface design to offer a clear foundation for the EA implementation stage.

4.2 COMPANY ORGANIZATION STRUCTURE

The Information Technology Centre (PTM) at Universiti Tun Hussein Onn Malaysia (UTHM) operates under a structured hierarchy led by a Director and a Senior Deputy Director, supported by an Office Secretary Assistant. The department is organized into six main divisions that handle different parts of the university's technology needs. The Application Management Division is responsible for the Application Development Unit and the Student and Staff Information System Unit. Infrastructure and Cyber Security handles data centre, networks, and telecommunications.

The Digitalization Technology, which handles web and multimedia services, and ICT Facilities and Support, which manages technical help and campus equipment. To ensure projects follow university rules, the Governance, Strategic, and ICT Training oversees planning and staff training. Finally, the Administration and Finance division manages the department's budget and office operations.

Figure 4.1 Organization Chart of Information Technology Centre (PTM) at Universiti Tun Hussein Onn Malaysia (UTHM)

4.3 CURRENT SYSTEM ANALYSIS

UTHM currently uses a combination of limited manual processes and computerized systems. Their digital operations are supported by a Tier 3 infrastructure. The data center uses Data Structure Infrastructure Management (DCIM) monitoring, air-based condenser cooling system and hot and cold aisle containment to track performance and reduce power consumption. Additionally, they have a staging area where vendors set up systems before entering the containment area.

Furthermore, UTHM's primary digital technologies include a central student database and Learning Management System (LMS) for teaching materials. These systems are not fully integrated so the majority of data transfers are carried out manually one at a time especially when the format or structure varies between the systems. However, API connections are used to manage big data transfers to increase efficiency and lower errors. The process was also carried out for data replication to the Data Recovery Center (DRC) where data is transmitted for disaster recovery to a DRC in Pagoh. Moreover, Single Sign-On (SSO) was implemented to make system access easier but frequent user concerns like forgotten password and Wi-Fi login issues still happen.

Additionally, UTHM is transitioning from old systems to a new web-based integrated system called Total Campus Integrated System (TCIS). This migration is ongoing and being completed phase by phase based on modules. Even though UTHM wants to be entirely digital by 2030, some administrative tasks are still done manually because of outdated equipment and legacy systems. They stated that some hardware comes with outdated operating systems which makes it less compatible with modern technologies. In addition, UTHM uses a WAF (Web Application Firewall) to secure web portals and websites as well as two firewalls which filter threats often from China and another from Europe to protect the system and data.

Overall, UTHM is actively pursuing digital transformation and has built a strong infrastructure. However, system integration, manual data handling, legacy equipment and user access continue to be challenges in achieving full end-to-end digital transformation by 2030.

4.4 SYSTEM REQUIREMENTS GATHERING

System requirements gathering is the process by which we gather data to understand user needs and the enterprise system that will be developed. It helps us to determine what functions, features and improvements are needed. Two methods were used to collect data for this industrial visit which are questionnaire and interview. Before visiting UTHM, a set of Enterprise Architecture related questions was prepared based on the TOGAF elements. Moreover, we asked the presenter directly during the talk session in order to gather more information. Below are the sample questions that were presented using the TOGAF elements:

1. Business Architecture

- What are the university's primary business capabilities (e.g., teaching, research, finance, HR)?
- What is the top 10 business processes that require transformation?
- Which processes are manual, duplicated, or inefficient in the current environment?
- What future state vision or strategic goals does UTHM want the new EA to support?
- What are the critical KPIs (student success, research productivity, operational efficiency)?

2. Data Architecture

- What core data entities exist (students, staff, courses, finance, research, facilities)?
- What data standards or naming conventions does UTHM currently use?
- Where is data duplicated across systems?
- Is there single master database in UTHM for student data or separate with student database and learning database?
- What current workflow transferring between student data to LMS from the master database?

3. Application Architecture

- What are the current enterprise systems in use (student system, finance system, LMS)
- What functions do these systems provide today?
- What functions are missing?
- What is the most common complain receive from student for current online portal?
- What is the desired future application landscape (modular, unified, cloud-based)?

4. Technology Architecture

- What is the current hosting model (on-premises, cloud, hybrid)?
- Which operating systems, databases, and middleware solutions are used?
- What hardware or infrastructure is aging or needs replacement?
- What is the preferred technology stack for the new system?
- What precaution steps will be taken if electric is cut off?

5. Integration / Inoperability Architecture

- What systems need to integrate with each other (student system ↔ LMS ↔ finance)?
- What integrations currently exist, and what issues do they have?
- What is the strategy for data migration?

6. Security Architecture

- What are the authentication requirements (SSO, MFA, LDAP/Active Directory)?

7. Information / Content Architecture

- How are files stored today (file servers, Google Drive, departmental systems)?

8. Infrastructure Architecture

- How many data centres does UTHM operate?
- What backup and recovery processes are in place?

9. Governance Architecture

- Who makes decisions about IT investments and architectures?
- Does the university need EA governance tools (e.g., architecture repository)?
- Who will be responsible for reviewing or approving the system?

10. Change Management & Implementation Architecture

- How ready are stakeholders for digital transformation?

- What departments are most resistant to change?

4.5 SYSTEM DESIGN

This section describes the design of the proposed system based on the requirements identified during the analysis phase. The system design is presented using enterprise architecture models, system architecture, UML diagrams, database design and interface design to clearly illustrate how the system is structured, user-friendly and capable of improving the efficiency of the existing processes at UTHM.

4.5.1 ENTERPRISE ARCHITECTURE EA – AS IS

Figure 4.2 illustrates the AS-IS enterprise architecture of UTHM. The figure was designed based on the requirements gathered during an interview session with their representatives. The architecture was designed according to their business goals, limitations and constraints specified by them during the interview. Each component of the EA will be discussed on the following sub chapters.

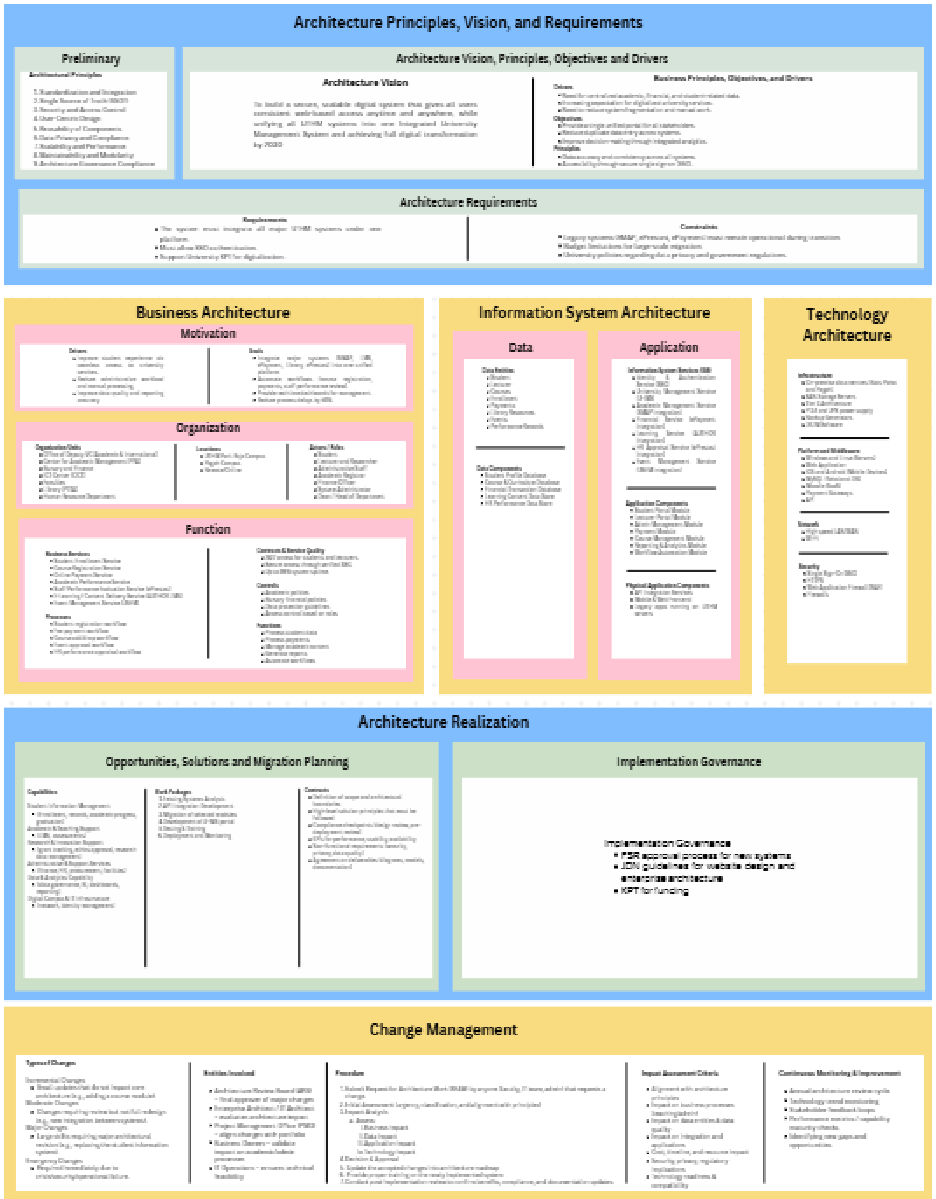

Figure 4.2: Enterprise Architecture AS-IS

4.5.2 COMPONENT EXPLANATION OF ENTERPRISE ARCHITECTURE EA – AS IS

Figure 4.3 shows the diagram that illustrating the architectural principles, visions and requirements. The primary goal of this design is to accomplish complete digital transformation by 2030. This is built on a goal to create a secure, scalable, and unified Integrated University Management System that offers all users uniform web-based access from anywhere. Several business reasons are driving this change, the most important of which is the necessity to combine distributed academic and financial data, as well as the growing demand for digital services. UTHM hopes to provide a single interface for all stakeholders, minimizing manual effort and boosting decision-making through integrated analytics.

The current Architecture Vision (AS-IS) is primarily driven by operational stability and department independence rather than a unified enterprise strategy. It focuses on maintaining the functionality of distinct legacy systems like SMAP and ePayment to ensure daily academic and administrative continuity, relying on established on-premises infrastructure and manual data synchronization to meet immediate, localized business needs without disrupting ongoing university operations.

The effective execution of this vision is dependent on achieving architecture requirements, such as integrating all key existing UTHM systems and implementing Single Sign-On (SSO) for smooth user authentication. However, these objectives must be balanced against real-world constraints. Developers must guarantee that essential legacy systems, such as SMAP, ePrestasi, and ePayment, remain fully functioning during the transition phase. Furthermore, the change must be completed within the constraints of the university's budget and tight government data protection requirements.

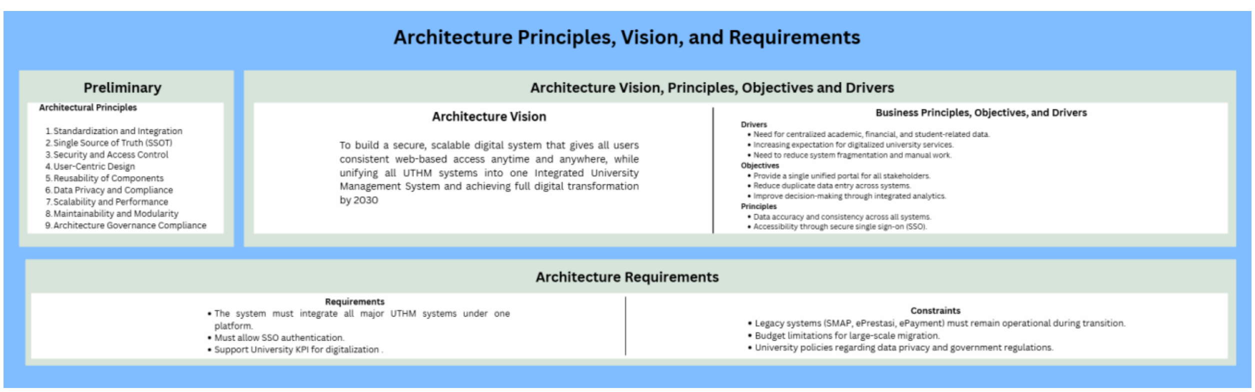

Figure 4.3 Preliminary and Architecture Vision element AS-IS

Figure 4.4 illustrates the business architecture, information systems architecture and the technological architecture in the EA. The business architecture layer is the foundation, outlining the motivation, organization, and function of the university's digital transformation. It is driven by its goal to improve student experiences while reducing administrative costs by integrating significant systems such as SMAP and LMS. An important performance aim is to cut process delays by 50% while maintaining 99% system uptime. This layer distinguishes between different organizational units, such as the ICT Center and the Bursary, and describes the responsibilities of individuals ranging from students and lecturers to system administrators. Functionally, it defines basic services such as student enrollment and fee collection, which are guided by academic and financial rules to ensure high standards of service.

This information system architecture defines the gap between business needs and technological execution by focusing on data and application components. Data architecture specifies the important entities such as students, lecturers and courses, in addition to the underlying data components, such as the student profile and financial transaction databases, which contain this information. The application architecture describes the Information System Services (ISS), which include the University Management Service (U-IMS) and particular modules for students and teachers. It focuses on the integration of historical systems (such as ePayment and ePrestasi) via an API integration layer, guaranteeing that disparate software components may efficiently communicate under a single site.

The Technology Architecture defines the physical and logical infrastructure needed to support the above levels. The infrastructure is created on-premises data centers in Batu Pahat and Pagoh, with Tier 3 architecture and SAN storage servers to provide high availability. The Platform and Middleware section includes information about Windows and Linux platforms, MySQL databases, and iOS and Android mobile compatibility. This complete stack is linked by high-speed LAN/WAN and Wi-Fi networks. The design protects university data with powerful security methods such as Web Application Firewalls (WAF), HTTPS encryption, and Single Sign-On (SSO).

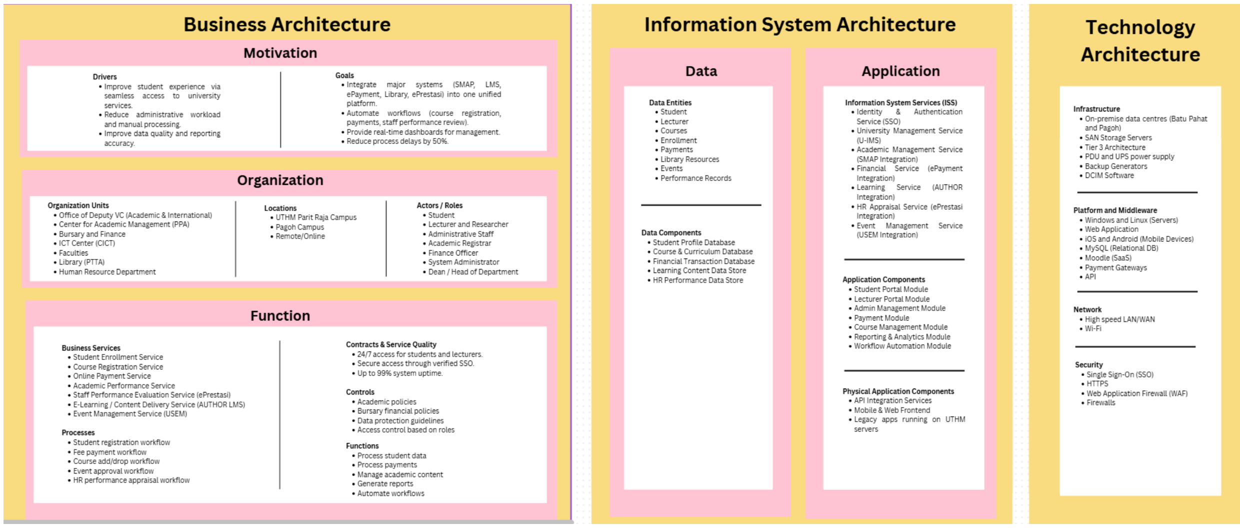

Figure 4.4 Business, Information System and Technology Architecture Element AS-IS

As shown in Figure 4.5, the current situation of UTHM's enterprise systems. At present, UTHM already has important capabilities such as Student Information Management, Academic Support, Administrative Services, data & Analytics, and IT Infrastructure. However, many of these systems still work separately. Some processes depend on manual work, and system integration is limited. Because of this, data sharing between systems can be slow, and it is harder for the university to monitor performance and make quick decisions. This situation creates an opportunity to improve efficiency and reduce system duplication.

From the migration planning perspective, the AI-IS stage focuses on basic analysis and preparation. The work packages mainly involve existing system analysis, initial API integration, and testing. These steps help UTHM understand which systems need improvement and how they can be connected. The contracts section highlights the importance of defining system boundaries, security requirements, and performance indicators before any major changes are made. Overall, the AS-IS architecture helps UTHM identify weaknesses in the current environment and prepares the university for a more structured transformation.

The Implementation Governance framework for UTHM's current enterprise architecture is defined by a structured oversight mechanism designed to ensure that all technical developments align with both internal policies and national standards. The governance process is anchored by three primary pillars which are the FSR approval process for the validation of new systems, the JDN (Jabatan Digital Negara) guidelines which standardize website design and enterprise architecture practices, and KPT (Ministry of Higher Education) for the strategic oversight of project funding. This framework ensures that any architectural changes, whether incremental or major, remain compliant with institutional mandates while maintaining the operational integrity of critical legacy systems.

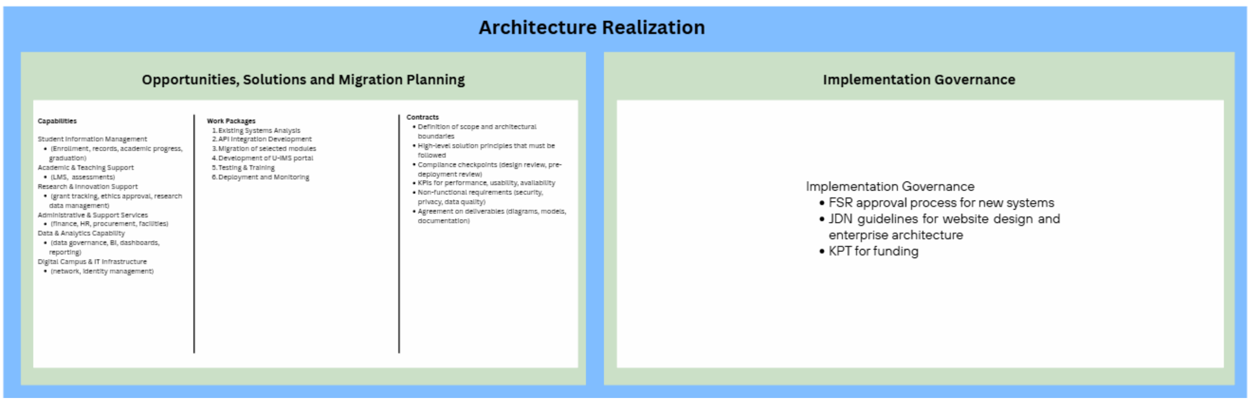

Figure 4.5 Opportunities and Solutions, Migration Planning and Implementation Governance element AS-IS

Figure 4.6 shows the change management component within the EA. The Architecture Change Management (Phase H) creates a governance structure to guarantee that the university's digital systems change in an organized way rather than through random changes. It divides modifications into three categories: process simplification, gradual process incrementation, and system redesign. This guarantees that minor enhancements are implemented quickly, while high-impact changes are examined with extensive evaluation required to preserve system reliability and security.

The Architecture Review Board (ARB) manages the process, which is started by a formal change request if there are new technologies or changes in business needs. Each proposal must go through an extensive impact analysis to determine its impact on data integrity, current integrations, and the user experience. This phase ensures that all technological evolutions stay firmly connected with the university's long-term strategic strategy for 2030 by requiring official permission prior to execution.

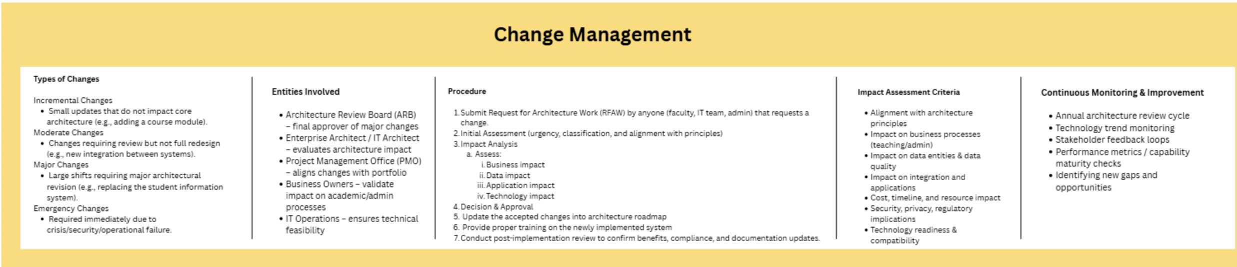

Figure 4.6 Change Management element AS-IS

4.5.3 ENTERPRISE ARCHITECTURE EA – TO BE

Figure x below shows the proposed TO-BE enterprise architecture. It serves as the blueprint for UTHM on the proposed recommendations that can be improved based on the AS-IS enterprise architecture that still aligns with their business while at the same time improving efficiency. Some components have changes based on recommendations to improve business model while some principles remain the same.

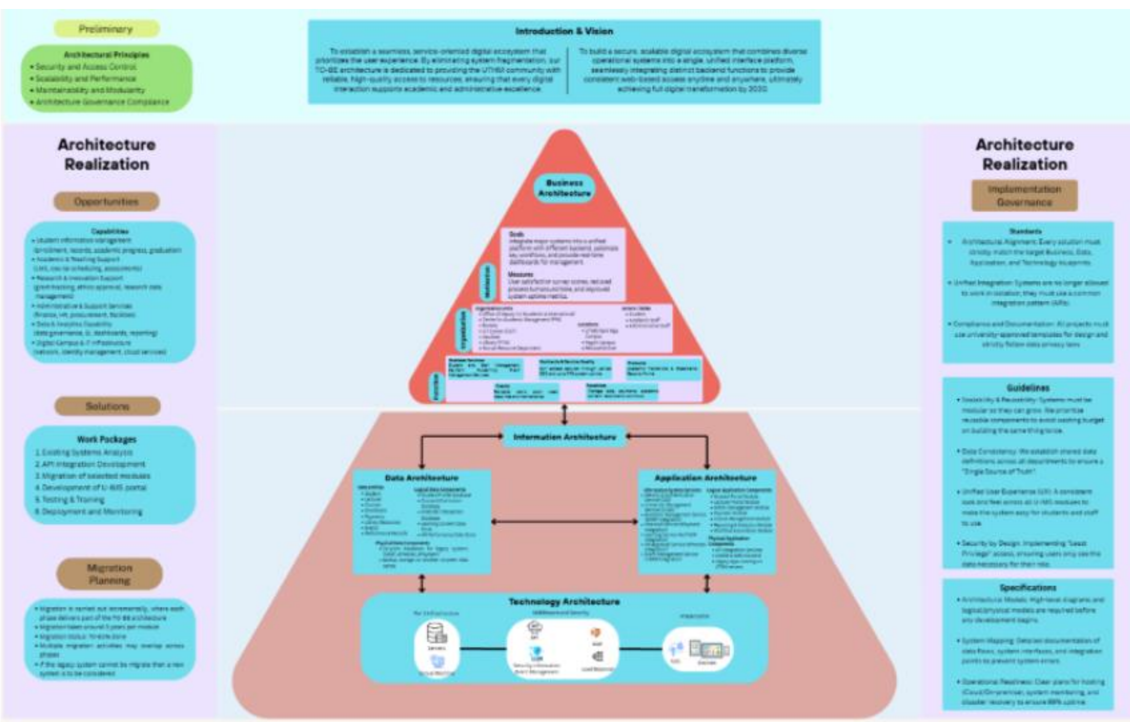

Figure 4.7: Enterprise Architecture_TO-BE

4.5.4 COMPONENT EXPLANATION OF ENTERPRISE ARCHITECTURE EA – TO BE

Figure 4.8 shows the diagram that illustrating the architectural principles, visions and requirements. The primary goal of this design is to accomplish complete digital transformation by 2030. This is built on a goal to create a secure, scalable, and unified Integrated University Management System that offers all users uniform web-based access from anywhere. Several business reasons are driving this change, the most important of which is the necessity to combine distributed academic and financial data, as well as the growing demand for digital services. UTHM hopes to provide a single interface for all stakeholders, minimizing manual effort and boosting decision-making through integrated analytics.

To ensure that the system has a strong structure, the preliminary phase develops four basic architectural concepts. These serve as the "base rules" for development, stressing scalability and modularity. The principles also focus on architecture governance compliance so that all systems follow the architecture planned. Besides, security is a non-negotiable component, with a particular emphasis on data privacy, compliance, and access control, to ensure the architecture stays controlled and legally sound.

The proposed Architecture Vision (TO-BE) aims to realize UTHM’s Digital Transformation 2030 agenda by establishing a unified, cloud-enabled Smart Campus ecosystem. This vision shifts the strategic focus towards high interoperability and a Single Source of Truth (SSOT), leveraging technologies like SAP Build Apps and SIEM to create a seamless, secure, and user-centric environment that integrates all academic and management functions into a single, scalable platform.

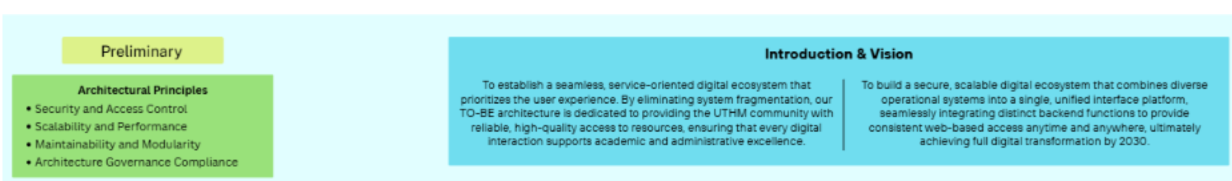

Figure 4.8 Preliminary and Architecture Vision element TO-BE

Figure 4.9 below shows the TO-BE business architecture, information architecture, and technology architecture.

The business architecture TO-BE remains the same as the business architecture AS-IS where it still focuses on improving the student experience and reducing the administrative workload. The architecture still involves the same organizational units and roles including academic, ICT, and administrative departments. In addition, the core function and services remain the same, such as Student Enrollment Service, Course Registration Service, Online Payment Service, and Academic Performance Service. This consistency that the existing business architecture is harmonized well with university's objective where it does not require a significant change in organizational structure or functionalities.

The TO-BE information system architecture represents a better future state than the AS-IS by moving from basic, siloed systems to an integrated, service-oriented environment. The TO-BE architecture centralizes and increases data entities such as enrollment, payments and performance records and adds standardized services like single sign-on and system integration which differ to the AS-IS architecture which primarily covers basic tasks with minimal integration. This change makes it possible for modular applications, enhanced security, improved interoperability, and increased scalability to meet academic and business requirements.

The changes from AS-IS to TO-BE technological design propose an increase in system durability, data flexibility, and continuous security monitoring. While the basic infrastructure remains focused on Tier 3 on-premises data centers in Batu Pahat and Pagoh, the TO-BE architecture incorporates offsite recovery to assure business continuity and data availability during local disasters, which is a significant improvement over the present system. Furthermore, the usage of VMware for virtualization results in a more resilient and scalable hosting environment than the present basic web application architecture. The network and security layers get major performance and monitoring improvements. The TO-BE architecture includes load balancers to manage large traffic levels. Most significantly, the security infrastructure is improved by the implementation of Security Information and Event Management (SIEM), which enables real-time threat detection and monitoring while supporting the university's current firewalls and Web Application Firewall (WAF).

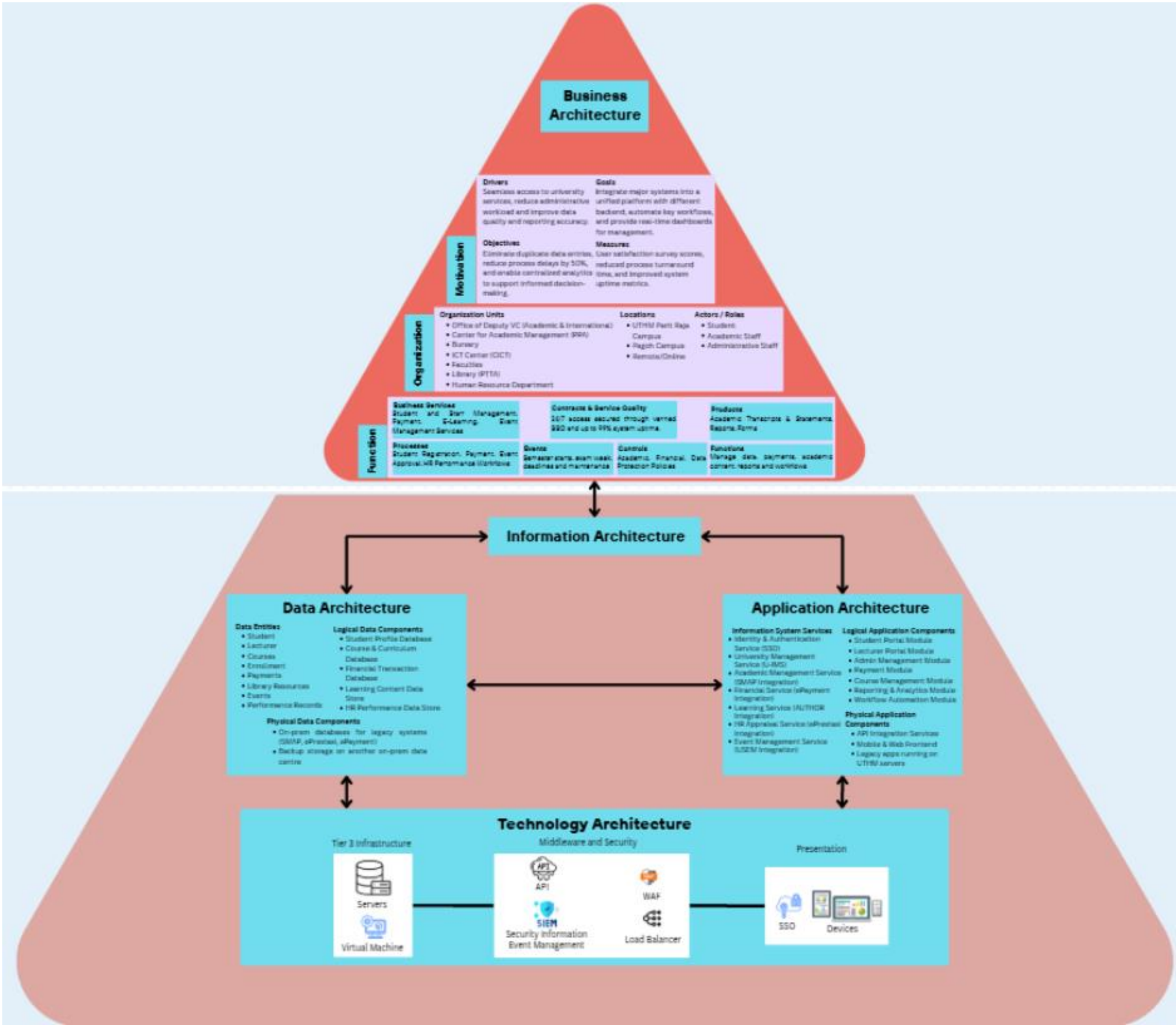

Figure 4.9 Business, Information System and Technology Architecture Element TO-BE

Figure 4.10 below presents the future direction of UTHM's enterprise systems. In this architecture, the focus is on modernization, integration, and digitalization. New solutions such as API integration development, and a unified U-IMS portal are introduced. These solutions aim to improve system connectivity, reduce manual work, and provide better user experience for students and staff. Data is managed more centrally, which helps improve reporting, data accuracy, and decision-making.

For migration planning, the TO-BE architecture outlines a clearer and more structured roadmap. The work packages include module migration, system development, testing, training, deployment, and monitoring. This ensures that systems changes are done step by step without disrupting daily university operations. The contracts section becomes more detailed by including compliance checkpoints, non-functional requirements such as security and privacy, and clear deliverables. Overall, the below figure shows how UTHM can move from a partially integrated environment to a fully digital, secure, and well-managed enterprise system that supports long-term growth and digital transformation.

Figure 4.10 illustrates the Implementation of Governance for the university's target enterprise architecture. It is structured into a comprehensive framework consisting of three core pillars which are standards, guidelines, and specifications. The standards section establishes the mandatory rules that all new IT developments must satisfy. Architectural alignment ensures every solution fits the university's master plan, while Unified Integration mandates the use of APIs to break down system silos. Furthermore, Compliance and Documentation standards ensure all projects follow legal data privacy requirements and use standardized design templates.

Furthermore, the Guidelines provide best practices for high quality system design. Scalability and Reusability promote modular build to save costs, and Data Consistency ensures a "Single Source of Truth" across all departments. To improve accessibility, the Unified Use Experience (UX) mandates a consistent interface, while Security by Design limits data access to a "Least Privilege" basis. Additionally, the specifications act as the technical blueprints required before launching. Architectural Models provide the visual structure of the system, while System Mapping documents precise data flows and integration points to prevent errors. Finally, Operational Readiness ensures that hosting, monitoring, and disaster recovery plans are in place to support the university's 99% uptime goal.

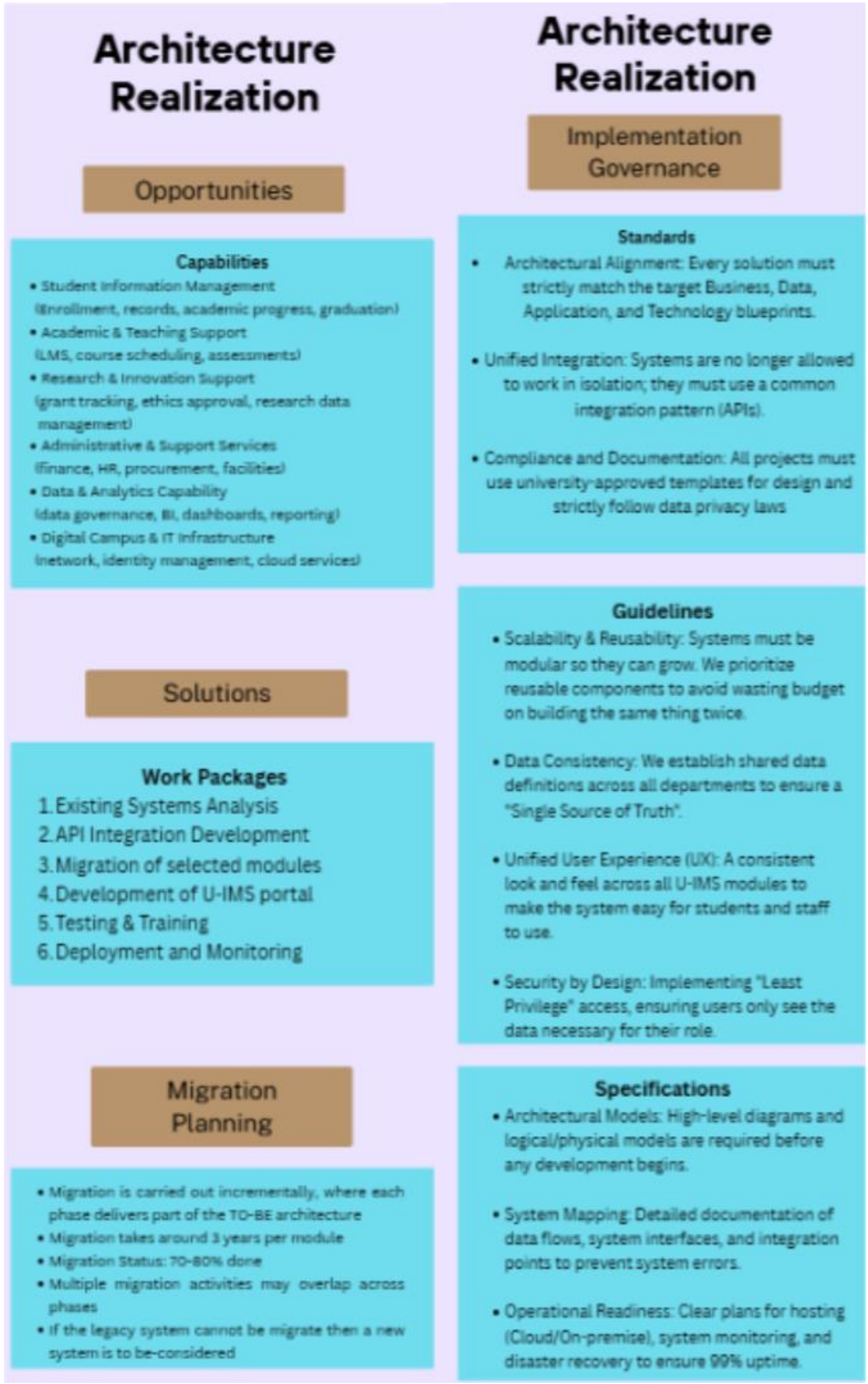

Figure 4.10 Opportunities and Solutions, Migration Planning and Implementation Governance element TO-BE

4.5.5 SYSTEM ARCHITECTURE

Figure 4.11 shows the cloud system architecture designed for the academic management platform. As the applications are developed using cloud platforms called SAP Build Apps and SAP Build Process Automation, a cloud-based architecture is used.

The academic management platform is hosted on SAP Cloud and built using SAP Build Apps and SAP Build Process Automation. The access to the system is provided to students, lecturers, and administrators through the Internet via web or mobile devices.

The core services including course registration, course management, section scheduling, admin dashboard, course approval, student profile management, and viewing student lists. These services are executed in the SAP Build Apps Runtime and SAP Build Process Automation Runtime.

The data is stored and managed using cloud-based, in-memory, and on-device storage, such as the app variables, page variables, and SAP Cloud databases to ensure fast and reliable access. The system access is protected through user authentication and role-based access control, while the system performance is monitored through application monitoring, user management functions and logging.

Overall, a secure and automated cloud-based solution for academic management is provided through this architecture.

Figure 4.11 Cloud System Architecture for the System

4.5.6 COMPONENT OF SYSTEM ARCHITECTURE

The three user's role which are Student, Lecturer, and the Admin can access the applications through their web browsers and mobile devices. Students use the system for registering for courses and viewing student profile. Lecturers access the system for viewing student lists and approving student course registration. Admins use the system for managing course and section and allocating time schedules and location for courses. This layer communicates with the backend through the Internet.

The Application layer shows the Services provided in the system. The services including registering courses and the approval process, viewing profile, and managing courses and sections details, class schedules, and class location.

The system uses NoSQL and in-memory database type. The main data component used are the app variable, page variable, and the on-device storage database. The app and page variables are temporary data stored when the application is running, while on-device storage database is a database that allows some data to be stored locally on user’s device when the application offline.

4.5.7 PROJECT DESIGN

The "Course Approval" subsystem of the Course Registration System is described in detail in this section.

4.5.7.1 USE CASE DIAGRAM FOR COURSE APPROVAL

Figure 4.12 shows the Use Case Diagram for the Course Approval Module, which is a functional subsystem of the Course Registration System. This module allows Academic Advisor to manage enrollment requests through viewing pending requests, approving course registration, and updating the status of each application.

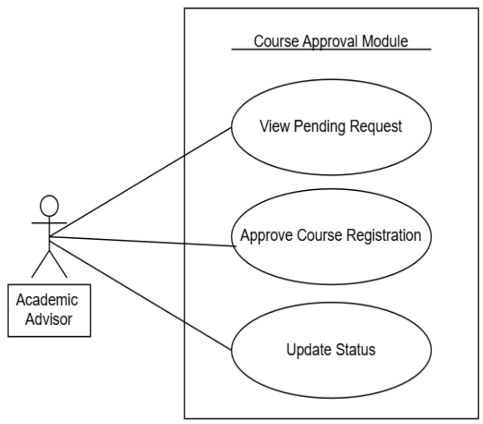

Figure 4.12 Use Case Diagram Course Approval

4.5.7.2 USE CASE DESCRIPTION

Table 4.1 user story description allows academic advisors to review course registration requests to ensure they meet faculty requirements.

Table 4.1: Use Case Description for Course Approval

Use Case: Course Approval
ID: UC007
User Story Description **As an** academic advisor, **I want** to review student registration requests, **So that** I can approve or reject them based on credit limits and section availability.
Flow of events: 1. The Academic Advisor logs into the portal and navigates to the "Approval Dashboard." 2. The system retrieves and displays the list of "Pending" registrations requests. 3. The Academic Advisor selects a specific student request to view details. 4. The Academic Advisor reviews the details and selects either "Approve" or "Reject". 5. The system prompts for a reason if "Reject" is selected. 6. The system updates the registration status via email.
Acceptance Criteria Postcondition : 1. The registration status is updated to "Approved" or "Rejected", and a notification email is sent to the student. Precondition : 1. The Academic Advisor must have valid staff's credentials, and the student must have submitted a request.

4.5.7.3 ACTIVITY DIAGRAM

Figure 4.13 illustrates the Activity Diagram for the Course Approval Module, showing the workflow between the Academic Advisor and the System. The process starts when the advisor views the pending list and selects a student's record. The system then checks if the request is valid. If it is valid, the status is updated to "Approved," but if it fails, the advisor must enter a reason before the request is "Rejected".

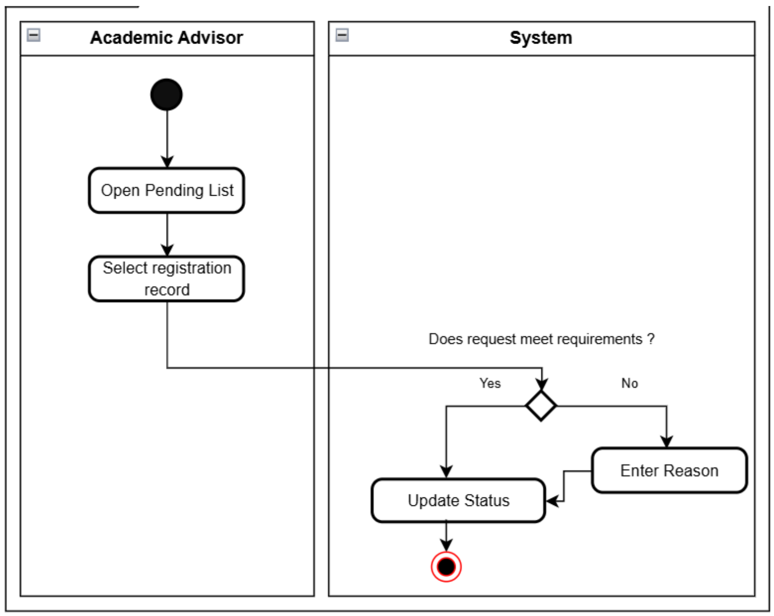

Figure 4.13 Activity Diagram Course Approval

4.5.7.4 SEQUENCE DIAGRAM

Figure 4.14 shows the Sequence Diagram for the Course Approval Module, illustrating the interaction between the Academic Advisor, the User Interface, the Database, and the Email Service. The workflow begins when the advisor triggers viewPending(), prompts the system to fetch and display a list of pending records from the database. Once a student is selected, the advisor decides through the processApproval() function, which follows an alternative logic in which the system either records an "Approved" status or requires a reasonMessage() for a "Rejected status". After that, a notification email will be sent to student email whether the status is "Approved" or "Rejected".

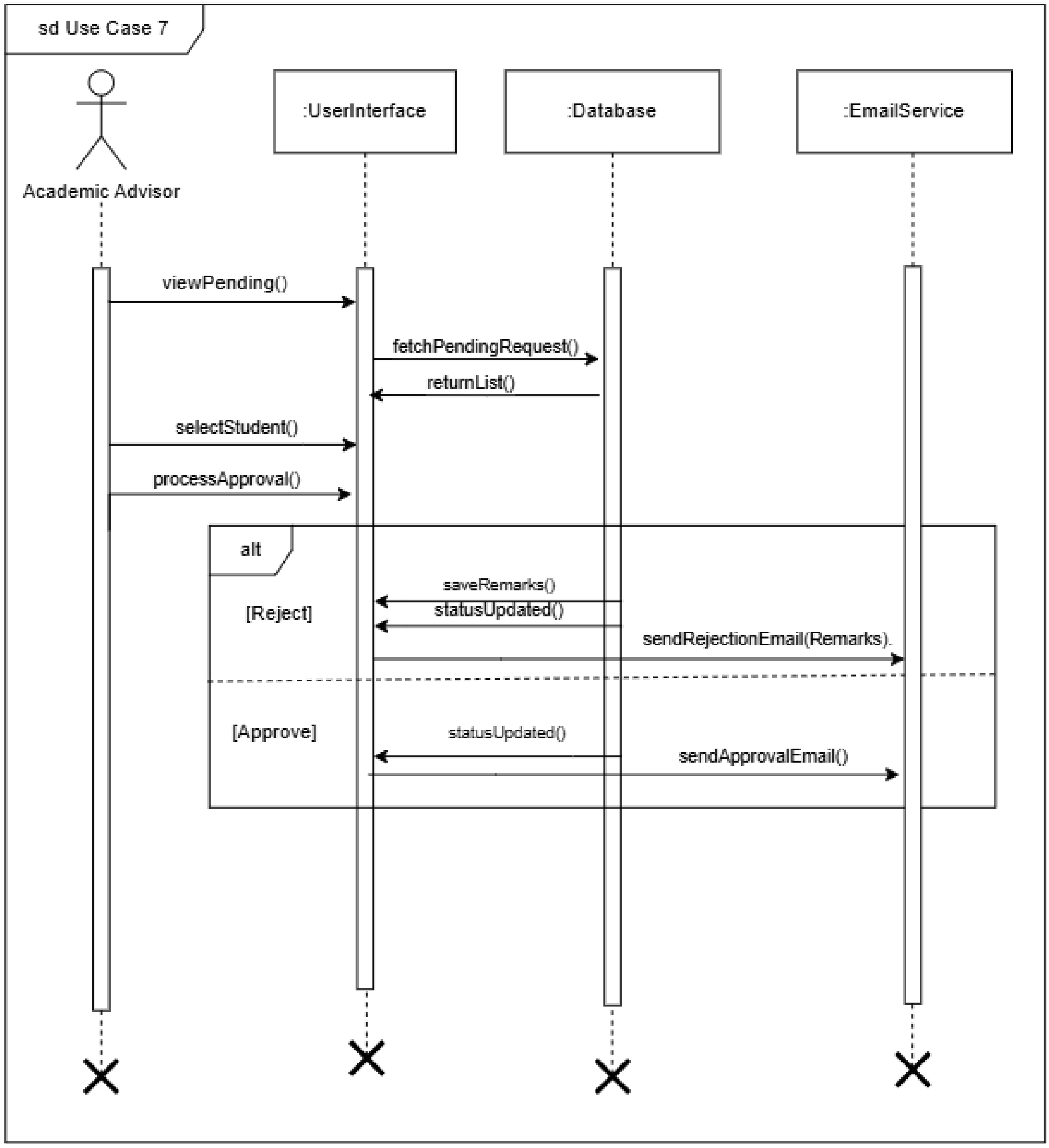

Figure 4.14 Sequence Diagram Course Approval

4.5.8 DATABASE DESIGN

Figure 4.15 shows the Entity Relationship Diagram (ERD), which defines the data structure for the Course Approval Module. The diagram shows a one-to-many relationship between the Academic Advisor and Course Registration entities, meaning one advisor is responsible for approving multiple student records. The Course Registration entity includes specific attributes such as regID, status, and remarks. This relationship is linked by the staffID which acts as a primary key in the Academic Advisor table and a foreign key in the Course Registration table.

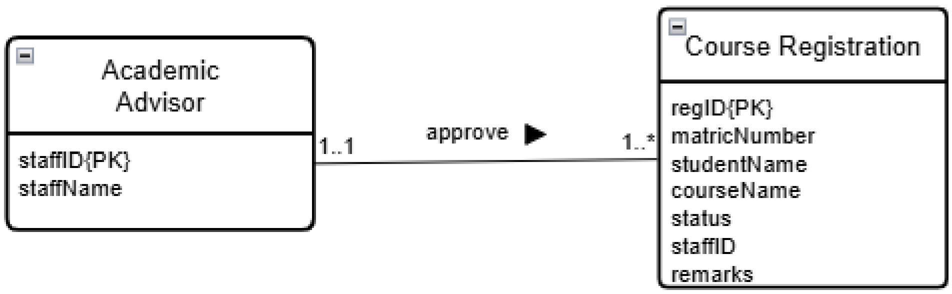

Figure 4.15 Database Design of Course Approval

4.5.9 INTERFACE DESIGN

Figure 4.16 illustrates the primary interface for student data entry. The form is structured to capture essential student information, including the student's name, matric number, and email address. A multi-select dropdown menu is provided for course selection, ensuring that all intended subjects are recorded accurately before the application is submitted to the workflow.

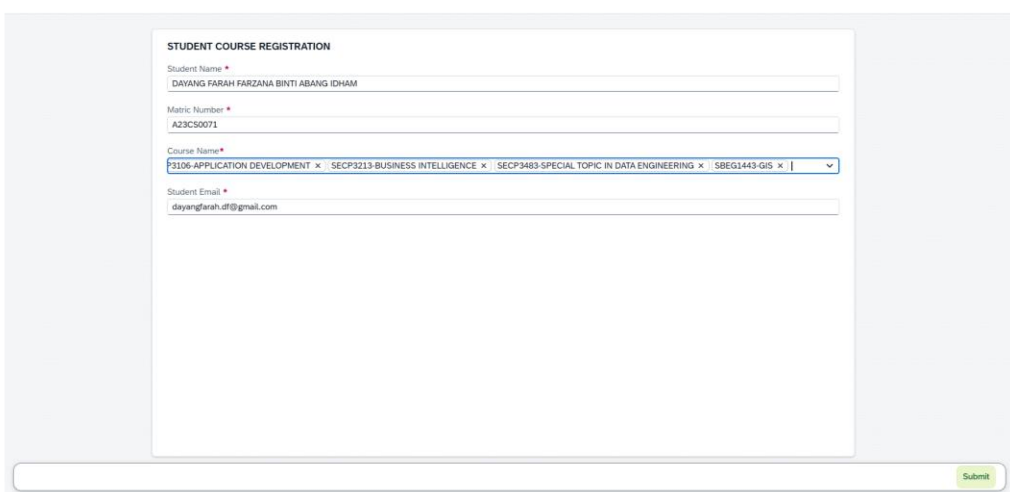

Figure 4.16: Student Course Registration Form

The Academic Advisor Approval Form is shown in Figure 4.17, which serves as the decision-making interface within the SAP My Inbox environment. All data previously submitted by the student is displayed in a read-only format to prevent unauthorized modifications during the review process. An input field is provided for "Advisor Remarks", allowing the Academic Advisor to enter justifications for the rejected application. The process is finalized by selecting either the "Approve" or "Reject" button located at the bottom of the interface.

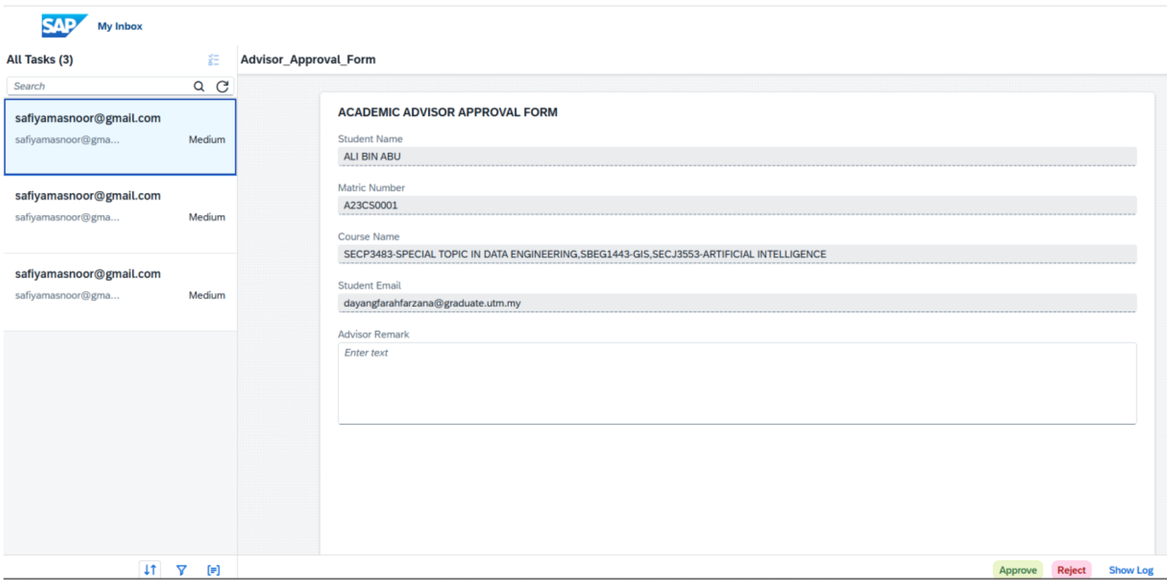

Figure 4.17: Academic Advisor Approval Form

Figure 4.18 and Figure 4.19 display the automated email notifications generated by the system upon completion of the advisor's review. In the event of rejection, the "Advisor Remark" is dynamically pulled into the message body to inform the student of the specific reason, such as the requirement to take additional subjects. Conversely, an approval notification is issued to confirm the successful course of registration of the student's application.

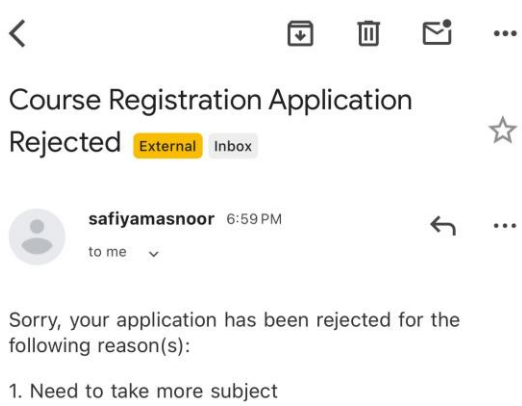

Figure 4.18 Rejection Email *Figure 4.19 Approval Email*

4.6 SUMMARY

This chapter has detailed the comprehensive analysis and design of the proposed Enterprise Architecture for Universiti Tun Hussein Onn (UTHM). The study began with an analysis of UTHM's organizational structure and current system environment, identifying critical challenges such as data silos, manual data transfer, and reliance on legacy infrastructure like SMAP and ePayment. The requirements gathering phase, conducted through interviews and site visits, highlighted the need for a unified platform to support the university's 20230 digital transformation vision.

Based on these findings, the chapter detailed the "AS-IS" architecture to document the existing gaps, followed by the proposed "TO-BE" architecture. The proposed design adopts the TOGAP framework to restructure the Business, Data, Application, and Technology layers, ensuring better alignment between IT capabilities and UTHM's strategic objectives. Key improvements include the introduction of a hybrid database model (combining SQL and NoSQL) to handle diverse academic data and the implementation of a robust Implementation Governance framework to standardize future IT developments. Ultimately, this chapter establishes the architectural blueprint required to transition UTHM towards a fully integrated, cloud-enabled Smart Campus environment.

CHAPTER 5
SYSTEM IMPLEMENTATION

5.1 INTRODUCTION

The analysis and design phase are a critical stage in developing the Course Registration Approval system using SAP Build Process Automation. This part focuses on defining the system, ensuring that the data captures from students flows logically to the Academic Advisor for review. By mapping out the entity relationships and process sequences, the design ensures that the system handles approval and rejections accurately. Furthermore, this phase establishes the technical requirements for automated notifications and database updates, which are essential for a reliable approval system.

5.2 SYSTEM DEVELOPMENT

This section details the technical implementation of the Course Registration Approval system using the SAP Build Process Automation platform. The development focuses on establishing a functional workflow that manages data flow from point of entry to the final notification stage.

5.2.1 PROCESS WORKFLOW AND LOGICAL BRANCHING

The system development phase involves the technical construction of the automated process and the configuration of communication protocols within the SAP Build Process Automation environment. This part ensures that the workflow logic is correctly implemented to handle data progression and email service.

Figure 5.1 illustrates the end-to-end logical sequence of the system. The process is initiated by the "Student_Registration_Form" trigger node. This component serves as the entry point for all student data, where the system ingests specific variables such as matric number, student, name, course selection, and student email. Once a submission is detected, the workflow engine instantiates a unique process ID, ensuring that the data is ready to be passed to the subsequent evaluation layer.

Following data ingestion, the workflow proceeds to the "Advisor_Approval_Form" which acts as the primary interface for the Academic Advisor. This node is configured to display the ingested student data in a read-only format, allowing the advisor to verify the request against academic requirements. Furthermore, the final structural component of the workflow is the logical branching that occurs immediately after the advisor's review. The system is programmed with a conditional split that differentiates between the "Approve" and "Reject" paths.

The conditional branching serves as the decision-making engine of the workflow, directing the process toward specific outcomes based on the Academic Advisor's evaluation. Within the

"Approve" path, the system is configured to trigger the "Approval Email" service, which sends formal confirmation to the student's registered email address. Conversely, if the request is routed through the "Reject" path, the system activates the "Send Mail" service. This path is designed to capture the specific remarks provided by the Advisor, ensuring that the student receives a clear justification for the rejection. Both branches eventually converge at the "End" node, which signifies the successful termination of the process instance and the completion of all automated tasks.

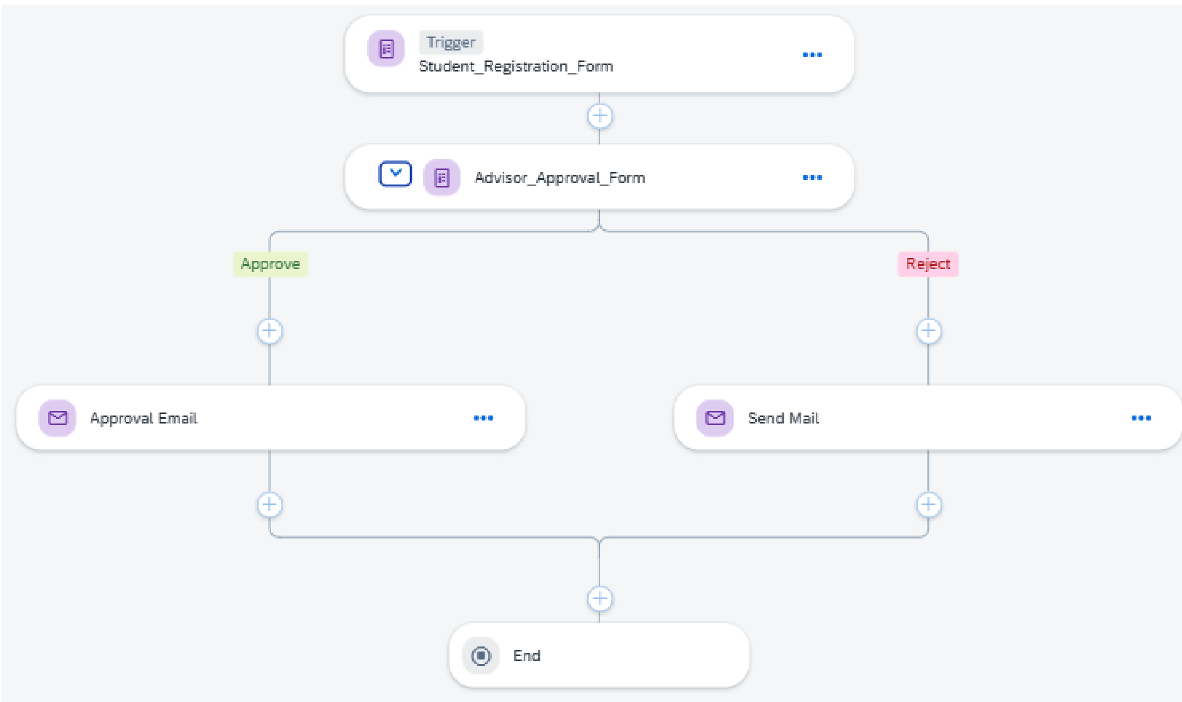

Figure 5.1: SAP Build Process Automation Workflow

5.2.2 SMTP SERVICE INTEGRATION

To facilitate automated communication, the system is integrated with a SMTP (Simple Mail Transfer Protocol) service, which handles the delivery of status notifications to the student. This integration represents the technical bridge between internal process logic and external communication.

Figure 5.2 shows the technical configuration of the automated email notification service. The service is programmed to dynamically retrieve the recipient's address from the initial data ingested during the trigger phase. Furthermore, the email body is configured to pull real-time data from the process variables, specifically the "Advisor Remarks" and the final "Status". By mapping this attribute within the SMTP settings as shown in figure 5.3, the system ensures that each student receives a personalized and accurate update regarding their course registration immediately after the advisor completes the review.

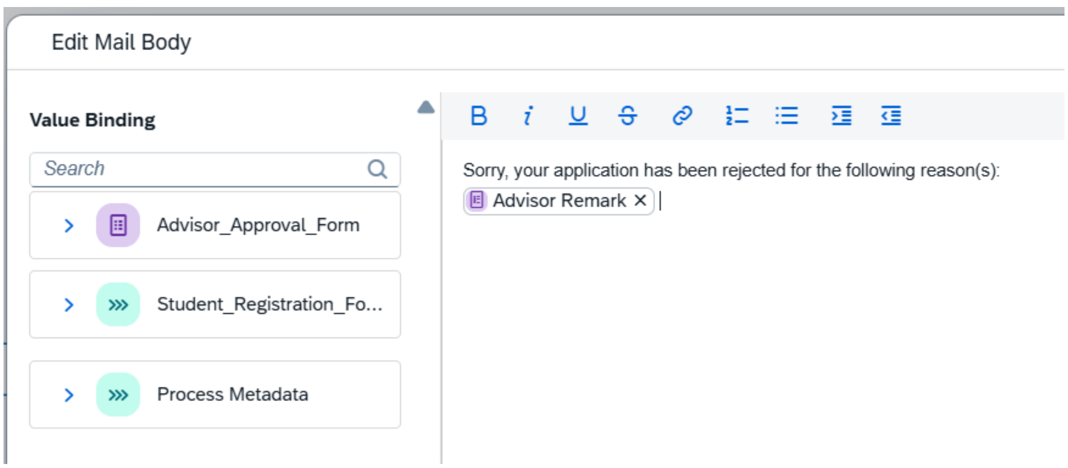

Figure 5.2 Configuration of Email Content

Figure 5.3 shows the setup for the email server that allows the system to send out notifications. The host and port settings establish the connection to the mail provider, while the authentication part ensures the system has the correct permission to send emails from the designated account. By having these settings correctly configured, the system can automatically deliver messages to students without any manual effort, ensuring the communication process is both fast and reliable.

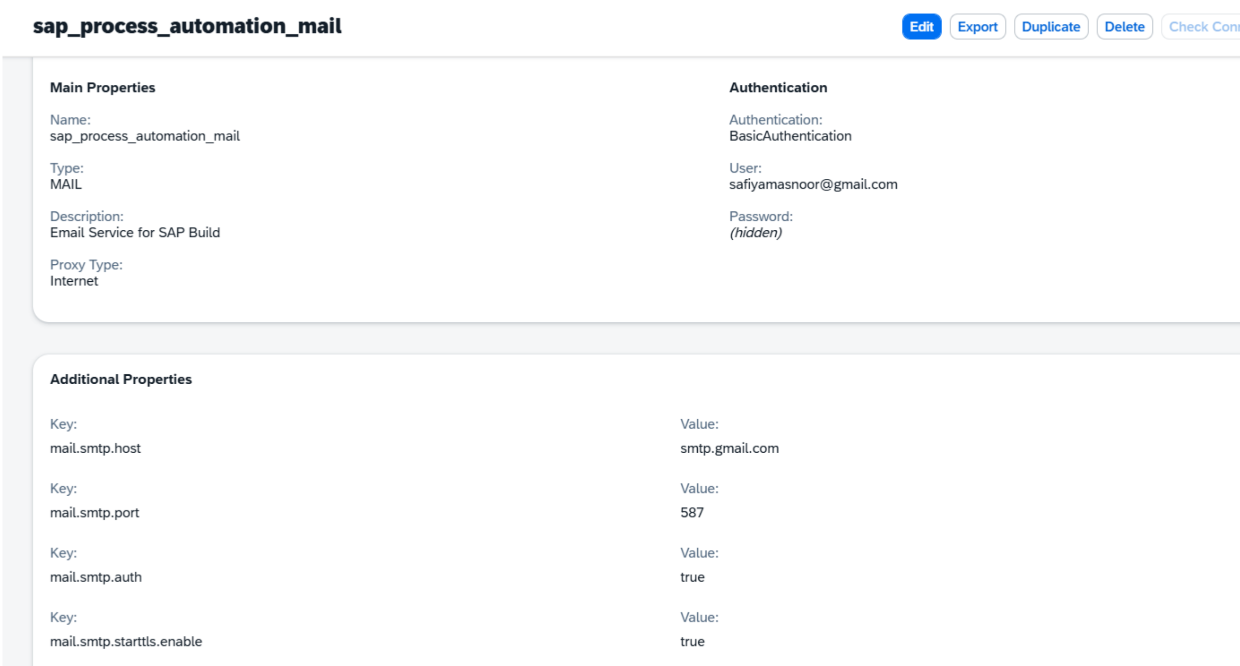

Figure 5.3 SMTP Service and Authentication Settings

5.3 DATABASE IMPLEMENTATION AND DATA BINDING

The implementation phase involves the technical configuration of database attributes within the SAP Build environment to ensure seamless data flow across the workflow. While the logical schema was established in Chapter 4, this part focuses on Data Binding, which serves as the technical link between the user interface and the system's internal data variables. By establishing these connections, the system ensures that information captured from students remains consistent and accessible throughout the approval process.

Figure 5.4 illustrates the technical mapping of form fields to the specific variables defined in the system's data schema. Through the Value Binding mechanism, each attribute such as studentName, matricNumber, courseName, studentEmail are linked to its corresponding input field. For instance, the data ingested from the Student_Registration_Form is mapped to the Advisor_Approval_Form inputs, allowing the Academic Advisor to review accurate student details in real-time.

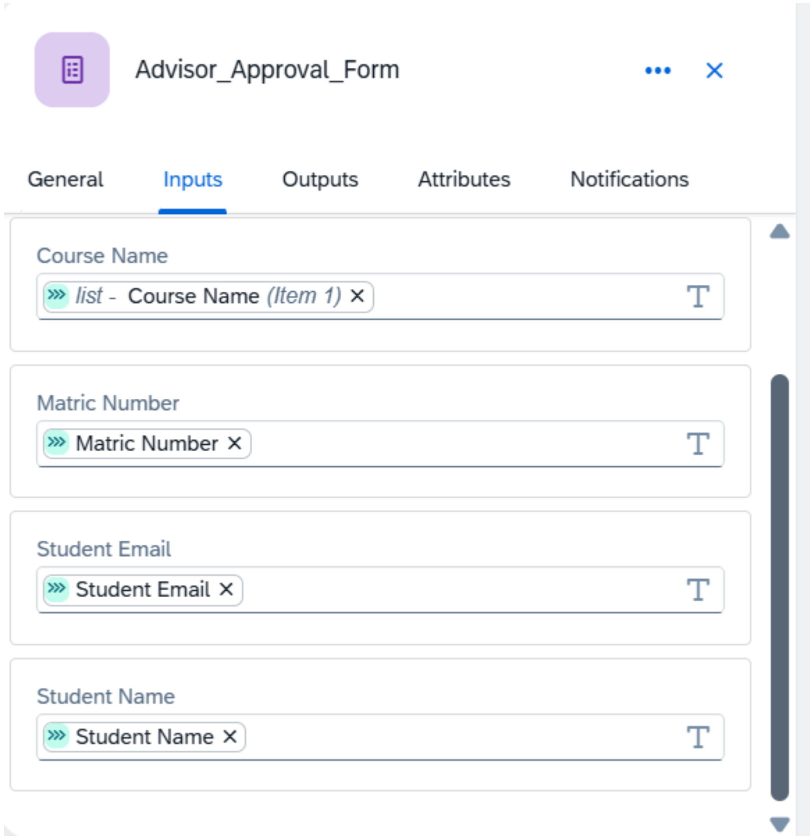

Figure 5.4 Data Binding and Attribute Mapping

5.4 CODING OF SYSTEM PROCESS

The coding of the system process is executed through the establishment of value-binding logic, where dynamic variables are mapped to automated service components to ensure that the workflow responds accurately to real-time inputs.

5.4.1 PROCESS DEFINITON AND VARIABLE LOGIC

The foundational logic is established by defining the global properties and input variables that the workflow is required to handle.

Figure 5.5 and Figure 5.6 show the configuration of the General Settings and Process Inputs within the SAP Build Process Automation environment. The process is formally identified as "Course_Registration_Approval_Process", and a specific data schema is programmed to recognize attributes such as Course Name, Matric Number, and Student Email. These variables are utilized as the system's primary data objects, ensuring that information captured at the trigger stage is preserved and accessible for all subsequent logical operations.

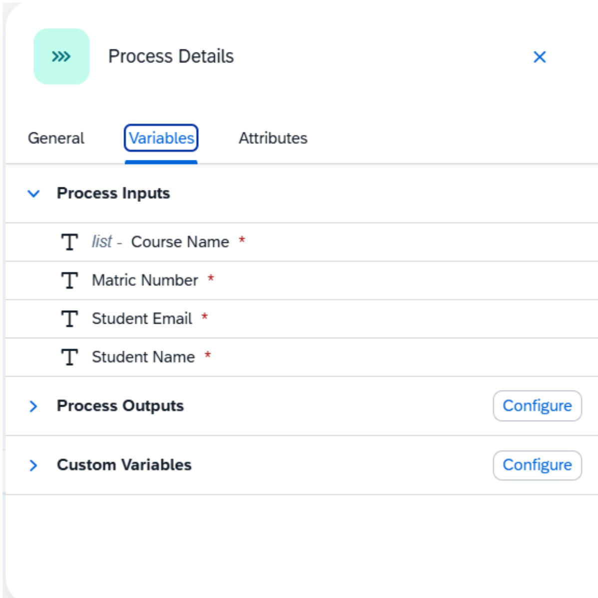

Figure 5.5 Input Variables

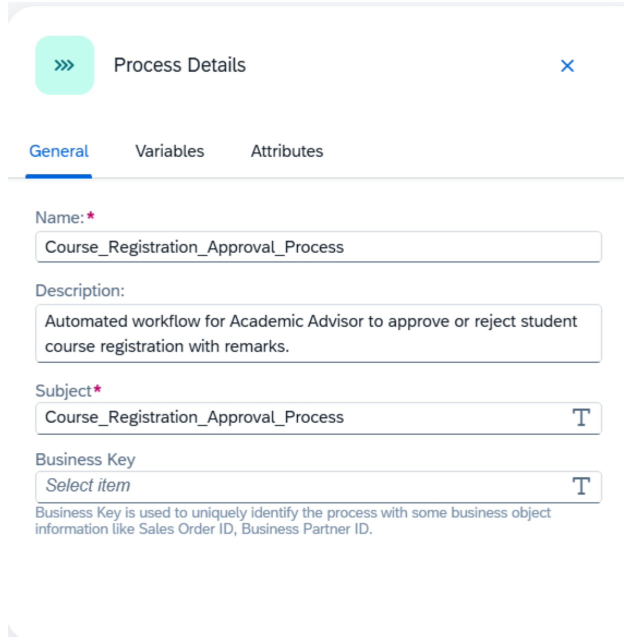

Figure 5.6 General Process Properties

5.4.2 LOGIC MAPPING FOR AUTOMATED NOTIFICATIONS

The functional intelligence of the system is finalized through the mapping dynamic variables to the automated email response components. Figure 5.7 illustrates the technical mapping applied to the "Send Email" service. The system is programmed to dynamically retrieve the recipient's email address by binding the "To" field to the Student Email variable. Furthermore, the emaik body is configured using Value Binding logic to automatically insert the specific "Advisor Remark" recorded during the evaluation phase. Through this configuration, a personalized and accurate notification is generated and delivered to the student immediately upon the advisor's decision.

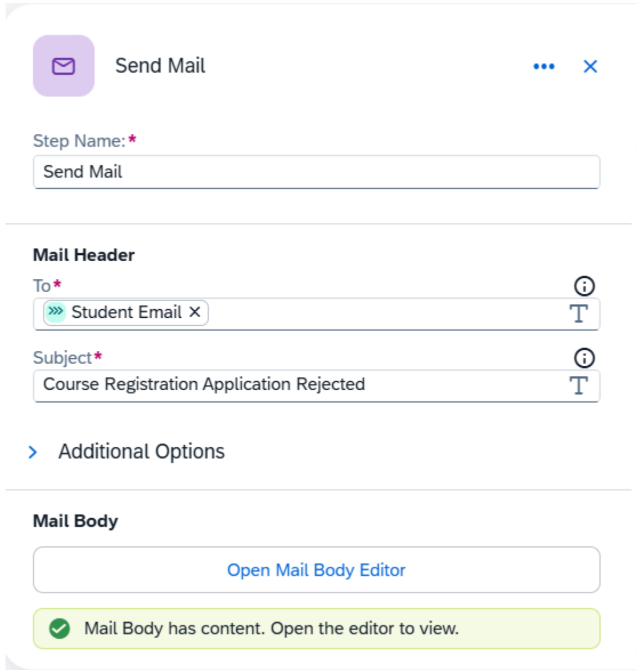

Figure 5.7 Mail Value Binding Logic

5.5 SUMMARY

In summary, the analysis and design phase has provided a comprehensive blueprint for the course registration system. The integration between logical database structure and the process workflow ensures that every student request is systematically recorded and evaluated by the Academic Advisor. Through the definition of clear decisions paths and automated email notifications, the design facilitates a consistent environment for the management of academic records. This structured approach ensures that the system remains scalable and maintainable while fulfilling all specified academic requirements within the SAP Build Process Automation platform.

CHAPTER 6
CONCLUSION

6.1 INTRODUCTION

This chapter provides a comprehensive review of the developed system, focusing on the evaluation of its functional performance, and its overall impact on student course registration. The specific roles played by each module in streamlining student course registration and administrative are examined to demonstrate the project's practical utility. Additionally, the technical constraints identified during the implementation phase are documented to provide a realistic assessment of the system's current capabilities within the SAP Build environment. The chapter concludes with strategic suggestions for future enhancements aimed at improving system integration and automation.

6.2 SYSTEM CONTRIBUTION

This project consists of seven modules, which are Student Course Registration module, Manage Course Credit Hour module, Section Scheduling module, Section Allocation module, Manage Student Profile module, Course Approval module, and View Student List for each Section module.

The Student Course Registration module is used to allow courses and sections to be selected and registered online. Besides, courses and sections can be deleted, and a real-time list of registered courses and sections can be viewed. The errors are minimized because real-time registered courses and sections can be viewed and confirmed before the course registration is submitted.

The Section Scheduling module is used to manage the venue and class time of each section. It is ensured that no sections clash with the class time or venue of other sections. The classroom usage is optimized and schedule clashes due to human error are prevented.

The Course Credit Hour Management module is used to manage course information. In this module, course can be added and managed by admins. Course management includes changes to the course code, course name, and credit hour. Moreover, courses can be deleted too.

The Section Allocation module is used to allow section requests to be viewed in the respective dashboard pages by administrators and students. Section requests can be submitted by students and viewed in the Student Dashboard. For admins, the section number can be updated based on the submitted request, and a list of submitted requests can be viewed in the Admin Dashboard.

The Admin Dashboard module is used to allow administrators to monitor and manage the course registration system. Registration statistics and analytics can be viewed on the main Dashboard page. For pending registrations, administrators can view the list of students awaiting Academic Advisor approval filtered by course. System settings such as registration period, credit limits, and notification preferences can be configured through the Settings page. Activity Logs are available to track all actions performed in the system.

The Course Approval module is used to allow registered courses to be reviewed and approved by academic advisers. It is ensured that registration data is transmitted accurately from the student to the academic advisor without manual handling. By guaranteeing that every registration is formally evaluated through a standardized workflow, academic control is maintained, and communication errors are minimized.

The Student Profile module is used to collect all student data including personal and academic information into one subsystem. Moreover, access to student data is supported for administrative purposes when needed. Student monitoring is improved, errors are minimized, and the redundant data entry is reduced.

6.3 SYSTEM CONSTRAINT

While the proposed architecture introduces significant operational improvements, several technical constraints were identified within the current design phase. A primary technical limitation is the system's dependency on stable internet connectivity. Since the solution is architected using the cloud-native capabilities of SAP Build Apps and remote NoSQL data storage, continuous network access is required for functionality. Consequently, latency or temporary inaccessibility to real-time scheduling data may be experienced in environments with limited bandwidth.

Furthermore, the integration with the university's existing legacy infrastructure presents a notable challenge. The transition from traditional relational databases –such as those used in SMAP and ePayment –to a modern NoSQL architecture necessity of complex data synchronization. Currently, this interoperability is proposed to be achieved through API bridges, which may impose limitations regarding data transfer speeds and format compatibility during the interim migration period.

FInally, regarding the development environment, the use of a Low-Code Development Platform (LCDP) was selected to facilitate rapid prototyping and proposal visualization. While this approach effectively demonstrates the system's core value and feasibility, the implementation of highly complex, customer algorithms –such as AI-driven automated conflict resolution –is bounded by the pre-defined flow functions available within the platform.

6.4 FUTURE SUGGESTION

The project has considerable amounts of opportunity to grow in the future, going from a number of individual modules to an integrated, professional system. The complete integration of the seven separate elements into a single ecosystem is the main plan for future development. Data input in the "Manage Student Profile" module does not now automatically appear in the "Student Course Registration" or "View Student List" modules due to the modules' current isolation. Through the use of SAP Business Technology Platform (BTP) to develop a centralized data backend, the system may provide a single "source of truth." This would eliminate data duplication and the possibility of human input mistakes by ensuring that every change is made in one module. For example, a student changing their credit hours is reflected across the whole system in real time.

The next suggestion is the implementation of process simplification and workflow automation. The system can be configured to have automated processes using technologies like SAP Build Process Automation. For instance, a submission in the "Student Course Registration" module could instantly alert a lecturer in the "Course Approval" module. This greatly reduces the workload for administrators by establishing a proactive system as opposed to a reactive one. Moreover, security requires the implementation of Role-Based Access Control (RBAC). This would guarantee that just registration and profile tools are visible to students, and administration modules such as "Section Scheduling" and "Section Allocation" are only accessible by those with authorization.

6.5 REFLECTION

The completion of this project has provided a profound understanding of how low-code development platforms, such as SAP Build, can be utilized to transform traditional manual processes into efficient, automated workflows within a formalized technological framework. The development of a proposed Enterprise Architecture (EA) for Universiti Tun Hussein Onn Malaysia (UTHM) underscores the critical necessity of aligning business processes with architectural blueprints, such as The Open Group Architecture Framework (TOGAF), to overcome existing operational silos. A key technical takeaway involved the importance of data integrity and logic mapping, as the successful execution of the course registration module depended entirely on the precise binding of variables to ensure a seamless data flow. Furthermore, the integration of automated SMTP services demonstrated how standardized governance can improve communication transparency by providing students with immediate, personalized feedback. Ultimately, this experience emphasized that a robust EA is not merely a technical requirement but a strategic asset essential for achieving UTHM’s long-term digital transformation goals, ensuring that future system enhancements are systematic, scalable, and user centric.

6.6 SUMMARY

In summary, the development of the automated course registration system successfully demonstrates the feasibility of using low-code technology to improve course registration workflows. While the system provides significant benefits in terms of data accuracy and communication transparency, it remains subject to technical dependencies such as internet connectivity and platform specific logic boundaries. The identified constraints and future suggestions provide a roadmap for transitioning this prototype into a more comprehensive, centralized ecosystem. Ultimately, the project fulfils its primary objective of modernizing the course registration process through automated approvals and secure data management.

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